Allianz Research | 25 March 2026

AI capex cycle: war-proof for now

Ludovic Subran
Chief Investment Officer and Chief Economist
ludovic.subran@allianz.com

Guillaume Dejean
Senior Sector Advisor
guillaume.dejean@allianz-trade.com

Alexander Hirt
Head Public Equity and Credit Global Investment Strategy
alexander.hirt@allianz.com

Katharina Utermoehl
Head of Thematic and Policy Research
katharina.utermoehl@allianz.com

Moritz Bartosch
Research Assistant
moritz.bartosch@allianz.com

Davide Corna
Research Assistant
davide.corna@allianz.com

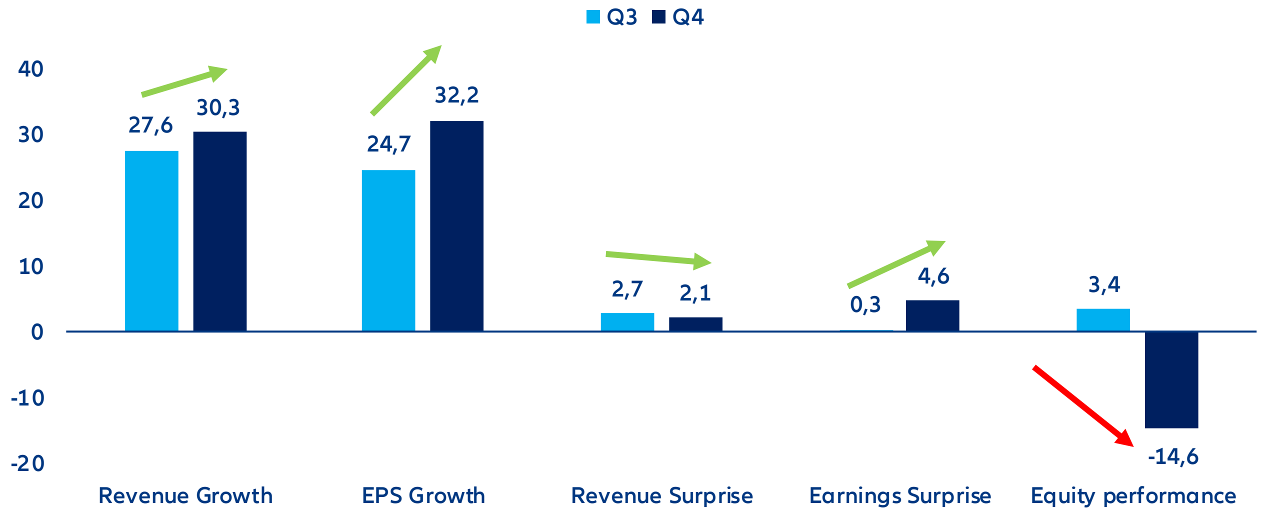
In Summary

- **AI hype deflates amid the capex-monetization debate.** Despite strong recent earnings, investor focus has shifted from profitability boost to revenue growth trajectory and cash-flow visibility, particularly given hyperscalers' elevated capex plans (≈USD 575bn, +50% expected in 2026) and weakening sentiment toward software amid risks of AI-driven revenue dilution. Our AI Bubble Risk Monitor continues to signal moderate bubble pressures: exuberant positioning has cooled, but widening credit spreads send signals of a higher market sensitivity over balance-sheet quality. Rising geopolitical tensions in the Middle East have further reinforced a rotation away from high-valuation tech as risk-off dynamics regain prominence.
- **The higher volatility regime will test the AI development regime, but the capex supercycle remains intact for now, supported by strong, counter-cyclical demand.** Technology capex has historically been sensitive to the macroeconomic environment and notably energy shocks as new inflationary pressure often results in higher interest rates, and consequently higher investment costs. In the US, tech is among the capital-intensive sectors most negatively impacted by severe energy price variations (~-30% correlation) but unlike energy, basic materials and utilities, the drop in investment does not result from windfall benefits stirred up by price effects. Nevertheless, hyperscalers' dominant market positions and substantial cash reserves reduce sensitivity to macroeconomic and geopolitical shocks. In parallel, public-sector investment is accelerating, driven by digital sovereignty and infrastructure build-out agendas. The data-center pipeline is robust (x2 at ~200GW by 2030) and momentum remains resilient despite geopolitical tensions, as illustrated by Germany's plan to double its capacity over the same horizon.
- **Energy volatility may reshape capex allocation rather than overall scale**. While near-term spending levels appear secure, the current concentration in data centers and cloud infrastructure could evolve. AI-related orders benefit from priority access within semiconductor supply chains, limiting immediate exposure to potential disruptions in South Korea and Taiwan linked to LNG and helium supply risks. However, a further ~50% increase in chip costs – as seen in Q1 2026 – could delay project timelines and accelerate the shift toward leasing models to share capital intensity (with estimated savings of 20–30%). At the same time, the case for expanding critical equipment and raw material production outside Asia is strengthening as current tensions expose the risks of supply-chain concentration and may rebalance investment away from the current bias toward software and computing services.
- **How are markets trading the "AI theme"? Focus is shifting from efficiency gains to revenue delivery.** While consensus broadly reflects current momentum, it is increasingly exposed to execution risk on both capex discipline and revenue realization. Valuations – mid-20s P/E for large caps – remain broadly justified but are vulnerable to short-term rotations in a high-volatility environment. The long-term outlook remains supportive, with capex anchored in tangible infrastructure orders that are partly decoupled from the pace of AI adoption. However, value creation is likely to be uneven across the technology stack, with relative winners in semiconductors and telecom equipment, and more challenged segments in software and consumer electronics.

AI cycle: capex monetization debate steals the show during last earnings season

The Q4 2025 earnings season confirmed the fundamental resilience and growth of the global technology sector. The NYSE FANG+ Index – our preferred gauge for tech and tech-enabled growth companies – showed robust financial performance in the most recent quarter. Aggregate results extended on previous strong delivery on expectations, with revenues rising by roughly +17% year-on-year and earnings by around +25%. Each of the leading tech firms exceeded estimates, on average by about 0.5% on sales and 1.6% on earnings across all constituents. Corporate guidance remained constructive, best highlighted by Meta's 5-10% upward revision to its sales outlook. Software companies also highlighted the fundamental potential of AI amid amplified concerns about the potential cannibalization of traditional service lines in areas such as legal, consulting and financial services.

Figure 1: Large-cap tech earnings season compared to past quarter (to end February i.e. pre-Iran war)



Note: Equity performance refers to the three-month period to the end of reporting season and relative to S&P Equal Weighted. Sources: Bloomberg, Allianz Research

Table 1: Earnings & revenue surprise and annual growth among listed US technology firms

	EPS			Revenue		
	Surprise	YoY%	Next quarter rev.	Surprise	YoY%	Next quarter rev.
Computer Hardware	7%	11%	0.0%	3%	7%	0.0%
Computer svs	8%	15%	-2.1%	1%	11%	0.0%
Consumer digital svs	11%	23%	-1.0%	1%	10%	0.0%
Consumer non-digital svs	1%	-11%	-9.4%	0%	15%	-0.4%
Consumer electronics	16%	33%	2.2%	4%	14%	1.7%
Electrical & Electronic Components	7%	32%	0.0%	3%	17%	0.0%
Electronic equip.	4%	8%	-0.1%	2%	8%	0.0%
Tech equip. manufacturing	6%	2%	1.9%	2%	8%	1.9%
B2B Services	5%	14%	-0.6%	1%	6%	0.0%
Semiconductor	6%	29%	0.1%	1%	16%	0.0%
Software	8%	20%	0.0%	2%	14%	0.1%
Telecommunications equip.	12%	19%	0.6%	2%	19%	0.4%
Telecom svs	1%	68%	0.0%	1%	2%	-0.1%
Transaction Processing Services	11%	22%	0.2%	2%	11%	0.2%

Data as of 10 March 2025 based on a sample of 383 companies. Sources: LSEG Datastream, Allianz Research

AI capex is set to continue increasing further in 2026, alongside investors' concerns. AI capex remains the defining market debate in 2026 as technology leaders race to build computing capacity they believe will secure – and deepen – their dominant position in a world where data and processing power are becoming strategic geopolitical assets. After rising ~60% in 2025, US Big Tech investment is expected to climb another +50%, surpassing USD600bn and lifting capex intensity to roughly 23% of revenue, more than double pre-ChatGPT levels. Tech now operates with higher capital intensity than telecom operators, whose investment ratios have declined as innovation shifts toward specialized satellite startups. Yet concerns are mounting: monetization timelines remain uncertain, power infrastructure expansion is lagging data-center demand (utilities capex +20% in 2025, +15% in 2026) and investors increasingly question funding massive spending cycles without the dividends or buybacks traditionally used to compensate capital-heavy business models.

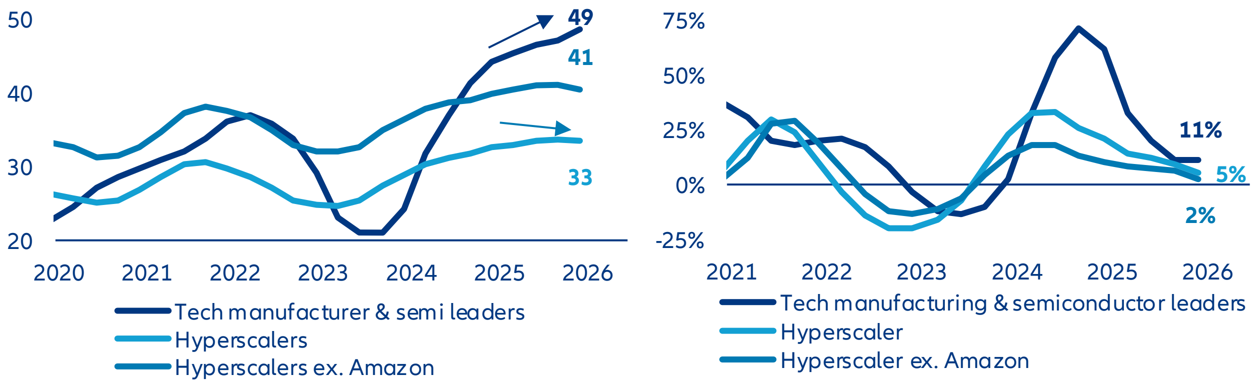
Table 2: Capex growth and intensity over revenue from top 50 biggest spenders within S&P 500 index, per industry breakdown

Industry	End-2023		End-2024		End-2025		End-2026	
	Capex ratio	YoY	Capex ratio	YoY	Capex ratio	YoY	Capex ratio	YoY
Technology*	11%	-4%	14%	44%	19%	58%	23%	50%
Telecom**	16%	-8%	15%	-4%	14%	0%	13%	-5%
Utility***	41%	11%	43%	4%	46%	20%	50%	15%

Sources: LSEG Datastream, Allianz Research

Markets are now questioning how long exceptional margins can last. Investors increasingly doubt the sustainability of profits boosted by quasi-monopolistic positioning and temporary demand–supply imbalances. Early signs from hyperscalers point to easing profit momentum, while Q4 2025 results – despite a +7% median earnings surprise and +17% annual growth – triggered no upward revisions to next-quarter estimates, reinforcing the idea of a profitability cap. Revenue growth remains solid but uninspiring (+2% median surprise vs. +12% for annual growth), reflecting expectations of gradual AI adoption constrained by regulation, costs, trust gaps and skill shortages. As margin expansion normalizes, markets are shifting attention to whether revenue acceleration can compensate, with IT and telecom services seen as most exposed to substitution risks and profit dilution.

Figure 2: Operating margin & profit-revenue growth gap of AI tech leaders* and hyperscalers



**Nvidia (chip design), ASML (chip equipment), TSMC (foundry), SK Hynix (memory chip manufacturer). Data as of Q4 2025. Sources: LSEG Datastream, Allianz Research*

Our [AI Bubble Risk Monitor](#) continues to hint at moderate bubble pressures, though the underlying drivers have changed to some extent. Fundamental developments around adoption momentum, data center vacancy rates and GPU product-cycle dynamics remain reassuring. Energy-market developments need watching but see no increase in risk. Meanwhile, exuberant expectations have deflated somewhat reducing the risk of an abrupt market setback triggered, for instance, by an earnings disappointment. In fact, equity positioning, the bear market indicator and 12m forward PE have all come down to more moderate levels over the past three months. Financial risks are on the rise – albeit moderately – and market signals are now flashing red when it comes to US investment grade tech spreads, which have recently moved to bubble-like territory, likely reflecting AI challenging software firms. But overall, bubble risks look less pronounced compared to a month ago. At this stage, tail risks over the long run remain contained but some shadow areas on the pace of capex monetization and rapid cash balance deterioration require monitoring, especially because capex is expected to continue increasing at a rapid pace this year, and because imbalances generated by this massive flow of funds could be broader than cashflow.

Table 3: AI Bubble Risk Monitor

	Indicator	Q1 2024	Q2 2024	Q3 2024	Q4 2024	Q1 2025	Q2 2025	Q3 2025	Q4 2025	Latest (Feb 25)
Market Signals	12M price return	2	1	1	1	0	0	0	0	1
	S&P 500 12M yoy breadth	0	0	0	0	0	0	0	0	0
	Data Center Spreads	0	0	0	0	0	0	0	0	0
	US IG Tech spread	0	0	0	0	0	0	0	0	0
	US HY Tech spread	0	0	0	0	0	0	0	0	0
	US HY Tech vs HY spread	0	0	0	0	0	0	0	0	1
	US IG Tech vs IG spread	0	0	0	0	0	0	1	2	3
	S&P 500 1Y implied vola	0	0	0	0	0	0	0	0	0
	10Y Treasury Yield	1	2	0	1	2	1	1	0	1
	S&P 500 Put/Call-ratio	0	0	0	0	1	0	0	0	0
Exuberant Expectations	Long-term earnings growth	0	0	0	0	0	0	0	0	0
	Earnings revisions (3M)	3	3	3	3	3	3	3	3	3
	Tech EPS forecast dispersion	0	0	0	0	0	0	0	1	1
	EQ positioning	1	0	0	1	0	0	1	2	0
	Bear market indicator	0	0	1	2	3	3	3	3	2
	12M forward PE	1	1	1	1	1	0	1	1	0
Financial Risks	Net-debt-to-equity ratio	0	0	0	0	0	0	0	0	0
	Cash on balance sheet as % of market value	0	1	1	1	1	1	1	1	1
	TED spread	0	0	0	0	0	0	0	0	0
	Cryptocurrency volatility	1	0	1	1	0	0	0	0	1
	Hyperscalers CapEx/Cash flow from operating activities	0	0	0	0	0	1	1	2	2
	US 5Y5Y Forward Expected Inflation	0	0	0	0	0	0	0	0	0
Fundamentals	Current Usage of AI	1	1	0	1	0	0	0	0	0
	Expected Usage of AI	2	1	0	0	0	0	0	0	0
	DC power demand vs generation supply growth							2	2	2
	Power capacity queues	1	1	1	1	1	1	1	1	1
	Electricity prices	0	0	0	0	0	0	0	0	0
	Operational Health	0	0	0	0	0	0	0	1	0
	Capex-to-sales level	1	1	1	1	2	2	2	2	2
	Data center pipeline (construction vs. commited)	0	0	0	0	0	0	0	0	0
	Data center vacancy	0	0	0	0	0	0	0	0	0
	Product cycle of GPUs	1	1	1	1	1	1	1	1	1

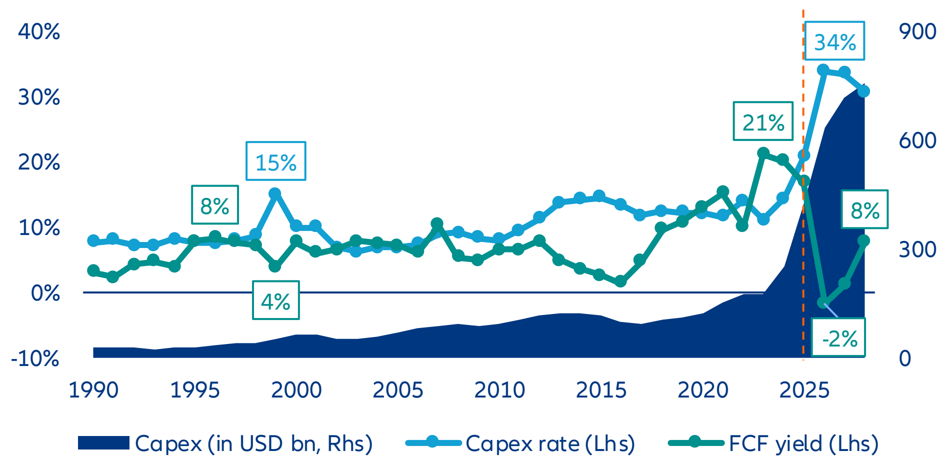
Legend: normal | elevated | overheated | bubble-like

Source: Allianz Research. Note: In addition to general market indicators, where applicable we also monitor AI sub-segments via equity baskets. The monitor assigns scores to each signal 0 = normal, 1 = elevated, 2 = overheated and 3 = bubble-like to help aggregate and compare diverse developments within and across dimensions. Thresholds for scores are derived statistically where long-time data series are available but also via expert judgement

Investment cycle: resilient to geopolitical tensions but watch for an allocation shift

Holding the capex intensity no matter what. The five largest US technology spenders are on course to commit over USD600bn in capital expenditure in 2026 alone – part of an estimated USD2.1trn deployment across 2026–2028 – driving capex intensity to 34% of revenue, more than double the 15% peak seen at the height of the 1990s internet buildout. Figure 3 tells the story starkly: absolute capex is now in vertical ascent, free cash flow is turning negative for the first time in 35 years and the inflection is sharper than anything in the modern history of listed US firms. Growth in capital spending will decelerate – from 51% in 2026 to 13% in 2027 and roughly 5% in 2028 – but that deceleration reflects the natural digestion of an extraordinary base, not a change of conviction. These firms are deliberately engineering a year of negative free cash flow because they have concluded that the revenue potential of AI justifies the sacrifice: technology penetration into enterprise and consumer segments is inherently gradual, and the returns on infrastructure laid today will accrue over years, not quarters.

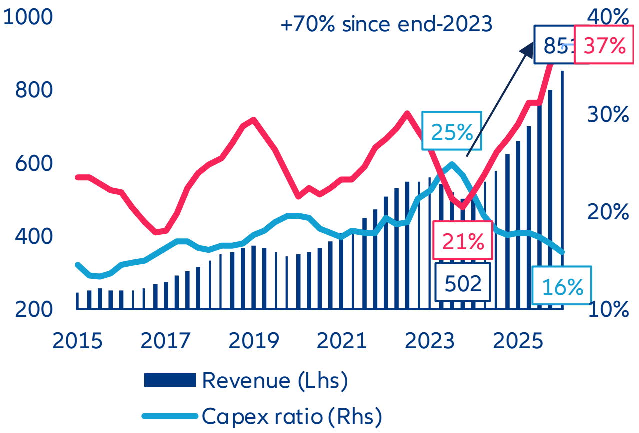
Figure 3: Capital expenditure (amount and intensity over revenue) and free-cash-flow yield of top five US capex contributors among listed firms

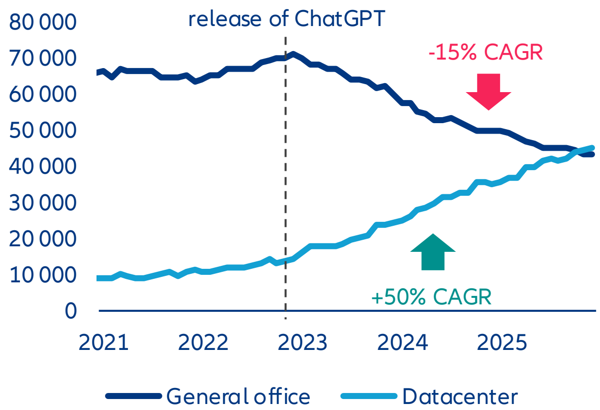


Note: The composition of top five capex leaders changes every year. Sources: LSEG Datastream, Allianz Research

AI benefits are already visible, but not yet broad-based. Hyperscalers are committing hundreds of billions of dollars to a technology cycle whose return horizon remains fiercely debated, and skepticism is rational. Yet, reducing this debate to a binary verdict – productive investment or reckless speculation – misses what the data already shows on the ground. US construction spending on data centers has exploded since 2023, while office construction continues its structural decline, a physical reallocation of capital that is reshaping the built economy in real time. Semiconductor revenues tell a similar story: Industry sales surged roughly 70% from end-2023, with operating margins expanding sharply even as capital expenditure intensity hit multi-decade highs. These are not paper gains – they reflect real backlogs, real pricing power and real profit flowing through the supply chain. The distribution of those gains is, admittedly, narrow: A handful of firms with defensible technology edges – in advanced logic, memory bandwidth, cooling and power infrastructure – are capturing a disproportionate share of the AI cake. But the notion that AI investment is generating only financial abstraction, widening inequality without creating tangible economic output, is countered by capacity-utilization rates, order books and the income statements of those positioned at the infrastructure layer. The more pressing question, then, is not whether AI capex is producing returns – it clearly is, for some – but whether those returns are being recycled in ways that build a genuinely durable and expansive ecosystem, or whether they are accruing narrowly and compounding existing structural asymmetries.

Figures 4 & 5 : Global semiconductor industry revenue, margin and capex intensity / Value of private construction spending in general office and datacenter in USA (annualized rate, in USD bn)

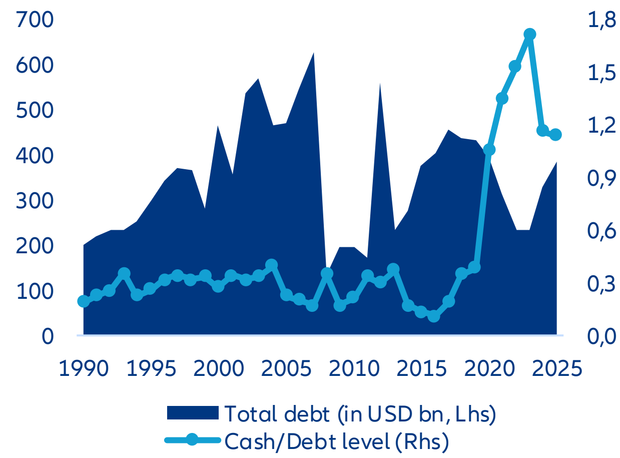


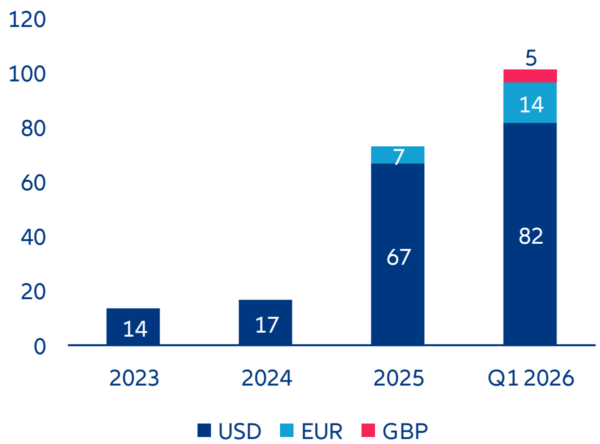


Right chart: November and December 2025 are preliminary data. Sources: US Census bureau, LSEG Datastream, Allianz Research

Solvency risks are overdone but rising debt still needs to be monitored. The parallel with heavy telecom investment in the late 1990s is imperfect as that was largely debt-funded and ended in widespread capital destruction; this round is being funded from the strongest corporate balance sheets in history, by firms with demonstrated monetization engines. But while solvency risks are not a major concern, we cannot overlook the ongoing shift of funding architecture and the increasing volume of AI-labeled debt since 2025. Even among hyperscalers who ignored the debt market for a decade, we note a structural pivot that is gaining more momentum in 2026 as they had already issued as of mid-March more debt than during the full year of 2025 (over USD100bn vs. ~USD80bn). With a total of USD385bn of debt reported in end-2025, the top US capex providers show a leverage ratio around -20% below their "high spender" peers in 2000, so ongoing rising leverage is not a matter of concern in the short term. However, the uptrend will be scrutinized by investors, alongside the downtrend on free-cash-flow.

Figures 6 & 7: Cash ratio & total debt level of top five largest US capex contributors / Corporate debt issuance of US hyperscalers* since 2023



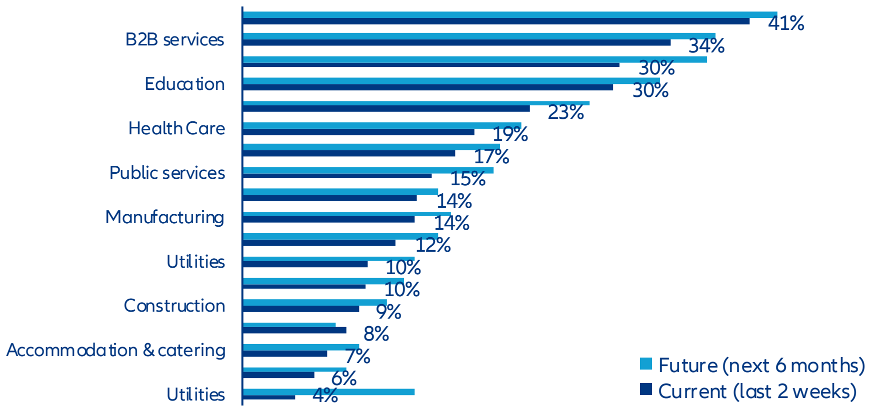


Left chart: data as of Q4 2025 / Right chart: data as of 18 March 2025. Sources: LSEG Datastream, Allianz Research

Beyond the noise: what actually shapes the AI capex cycle. The debate around AI investment has been too often reduced to market valuation stories – a narrative of soaring multiples, crowded trades and the occasional correction fueling breathless commentary about bubbles and busts. The more consequential discussion is industrial, structural and slower-moving. Four fault lines are quietly shaping how – and whether – current levels of AI capital expenditure can be sustained and productively absorbed. First, the pace of corporate and consumer adoption, which is tracking a natural learning curve rather than the vertical inflection that capital deployment implicitly assumes – a structural lag with direct consequences for return horizons. Second, a growing concern around misallocation: the breakneck pace of AI infrastructure buildout is not matched by equivalent investment in the underlying industrial capacity – power generation, grid infrastructure, advanced manufacturing – needed to underpin it, creating tangible shortage risks and inflating input costs across the supply chain. Third, a question of circularity: whether AI profits are genuinely creating new economic value or merely consolidating the dominance of a handful of firms, raising the specter of systemic spillover and contagion risks should any major actor stumble. Finally, new tensions in the Middle East, where energy supply disruptions could reignite inflationary pressure, complicate the rate outlook and dent the investor confidence on which this investment cycle critically depends. Taken together, these fault lines do not make the AI capex story one of inevitable disappointment but they do make it one that demands far more industrial rigor and strategic honesty than the market narrative has so far supplied.

1) **Gradual adoption is fueling monetization debate, not calling into questions capex itself**. The professional adoption of AI remains gradual, exploratory and uneven – and that is not necessarily a bad signal. Corporates broadly recognize AI's long-term potential, yet the transition is proceeding cautiously for structurally sound reasons: firms still need clearer visibility on implementation costs, measurable productivity gains, cybersecurity risks and an evolving regulatory framework before committing to large-scale deployment. Psychological barriers compound the technical ones as companies remain reluctant to build deep dependency on providers lacking long operational track records. The result is that most organizations are currently engaged in incremental process improvements rather than transformative reinvention – a pattern entirely consistent with how major technological shifts have always unfolded. Early gains are predictably concentrated in digitally mature sectors such as consulting, finance and insurance, while construction, utilities or agrifood show only limited impact so far. Financial transfer costs further widen this gap, with adoption among very small enterprises running nearly 10pps behind large corporates. Rather than the binary outcome between automation and employment destruction that dominates media narratives, AI is more likely to act as a complementary technology enhancing existing tasks and business models – with capex implications already shifting visibly toward staff training, compliance, cybersecurity and data infrastructure, the unglamorous but essential prerequisites for durable migration. For technology companies, this pace differential between capital deployed and returns materializing is not merely an investor-relations inconvenience – it is a strategic constraint that demands greater investment discipline and a genuine rethink of how financial risk is shared across the ecosystem. A measured adoption curve is not evidence of deterrence; it is the bedrock on which a solid and sustainable industry gets built.

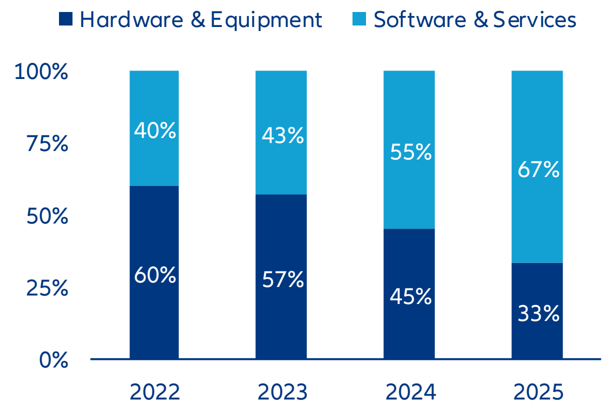
Figure 8: Survey of US firms about their current and expected use of AI technology, per sector breakdown



Data as of 22 February 2026. Sources: US Census bureau (biweekly survey on business trends and outlook), Allianz Research

2) **AI cannibalization of global supply raises question on industrial capacity capex lag**. A disproportionate share of global supply is being absorbed by the AI value chain – particularly for GPUs, high-bandwidth memory, advanced substrates, power electronics, silicon wafers and electronic manufacturing equipment. The result is tightening availability and rising input costs for other sectors. Memory chip prices, for instance, have more than doubled since last summer (see Figure 8) as limited fabrication capacity has been redirected almost exclusively toward AI-oriented demand, creating pass-through inflation for downstream customers and, in some cases, tangible shortage risks. Consumer electronics manufacturers, robotics and IoT producers, and especially the automotive industry – where software and semiconductor content per vehicle has surged in recent years – appear particularly exposed to these side effects. Higher component costs would compress industrial margins and/or raise final retail prices, potentially dampening end-demand for tech hardware. We estimate that manufacturers would have to increase retail prices by +7% for smartphones and +15% for premium laptops to protect half of current profit margins in case of doubling memory chip input costs. In turn, elevated equipment premiums may slow consumer adoption cycles, paradoxically constraining the diffusion of AI-enabled devices. While current AI usage among households remains largely concentrated in writing and search applications, scaling the technology into more embedded, hardware-intensive use cases requires affordable, dedicated equipment; inflationary pressures along the supply chain risk delaying that broader penetration and limiting the real economy spillovers that AI proponents anticipate. If we observe some lag in the capex programs of ICT equipment manufacturers, most of the capital is being captured by data-center development programs as industrial priorities are primarily focused on computing-capacity expansion (current 33%/67% allocation of capex between ICT hardware & manufacturing and ICT services – see Figure 9). Over the next two years, we see a change of capex structure and a stronger allocation of technology-related investment funds into new ICT hardware and equipment industrial capacity, narrowing the capex structure gap with digital & IT services toward a 50/50 ratio, like in 2024.

Figures 9 & 10: Memory chip spot price in USD (Flash NAND 64GB) / Capex distribution between technology hardware & equipment manufacturers and digital & software services providers

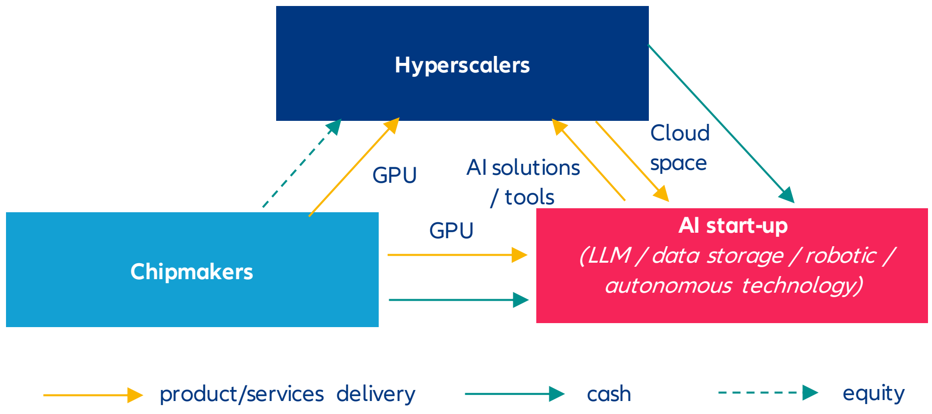


Sources: LSEG Datastream, Allianz Research

3) **Capital circularity casts doubts about the productive role of capex**. In the meantime, circular capital is increasingly dominating the AI ecosystem, especially across chips and cloud, creating a powerful – but potentially fragile – perception of strength along the entire supply chain. Semiconductor leaders such as Nvidia, TSMC or ASML sit at the core of this expansion, riding a +60% CAGR industry revenue surge since end-2022 and benefiting from multi-billion-dollar hyperscaler contracts, 30%+ CAGR cash accumulation and a global market revenue that could hit a USD1trn milestone this year if strong demand remains in check. Yet, beneath the surface, inventory growth among the top US chipmakers (+80% annually) is outpacing already explosive revenue gains, while part of demand stems from structurally unprofitable AI players reliant on external funding, raising questions about the durability of the cycle.
More concerning is that this revenue boom has not translated into proportional industrial reinvestment: capex intensity among the top five global foundries slowed last year to a five-year low at 24% (and hardly 2% for the top three US chipmakers), or almost 10pps below its 2023 peak. In the meantime, capex intensity

among top US designers stayed relatively flat at 2%, with profits being distributed to shareholders, invested in start-ups in early stages of development or directly to business partners to fund new data-center infrastructure. With hyperscalers and tech service providers now absorbing roughly two-thirds of total AI spending, capital allocation appears heavily skewed toward data-center infrastructure, exposing a systemic imbalance. Behind circularity stories, there is a deeper issue within the AI ecosystem: a strong capital interconnection between each key actor of the chain whose funding has been gradually outsourced to financial markets and private investors. That holds seamlessly as long as the AI monetization narrative holds. The moment investors seriously question the timeline – and the friction-heavy reality of enterprise technology adoption gives them every reason to – liquidity assumptions could crack across cash, credit and equity simultaneously, with spillover effects that would reach well beyond the technology sector.

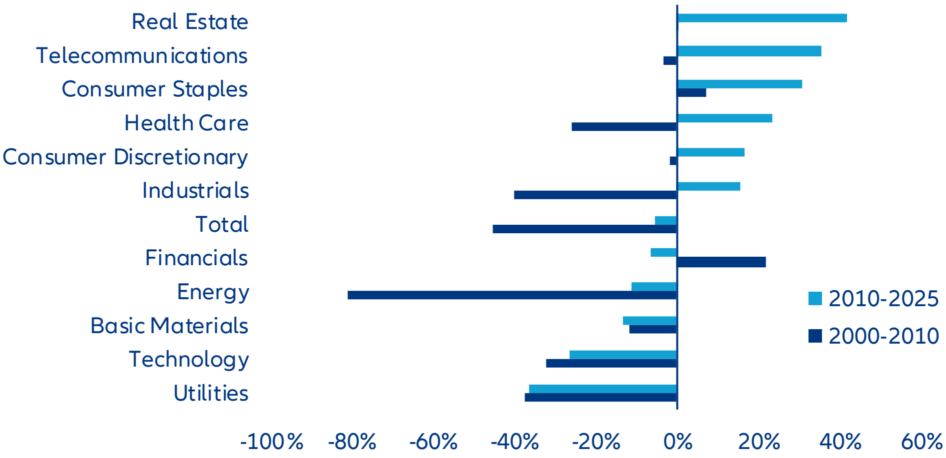
Figure 11: Circularity of investments from the AI ecosystem



Source: Allianz Research

4) **Middle-East tensions revive bad memories: a double chokepoint to monitor.** The re-escalation of Middle East tensions introduces two distinct but mutually reinforcing pressure points on the AI investment cycle – one financial, one industrial – neither of which the market has fully priced in. On the funding side, an energy-driven inflationary shock would complicate the rate outlook at precisely the wrong moment. Under already-intense scrutiny over capex monetization, higher-for-longer rates would give investors and lenders legitimate cover to tighten standards, demand stronger return guarantees and quietly stall the most speculative infrastructure projects. Given the extreme capital intensity of AI – where a single hyperscaler data center can absorb several billion dollars before generating a dollar of revenue – even a modest repricing of risk appetite could trigger meaningful investment pauses across the pipeline. This is not a theoretical sensitivity: the correlation between energy price volatility and technology sector capex in the US is among the strongest across all S&P 500 industries over both the 2000–2010 and 2010–2025 periods, suggesting that energy shocks reliably translate into investment hesitation across the tech supply chain. The silver lining, if there is one, is that capital discipline tends to accelerate efficiency: constrained funding historically pushes R&D toward leaner architectures, and the current cycle would likely see inference optimization and cost-efficient model deployment prioritized over brute-force training at scale.

Figure 12 : Correlation between capex and energy prices among US firms, per industry breakdown (S&P 500 benchmark)

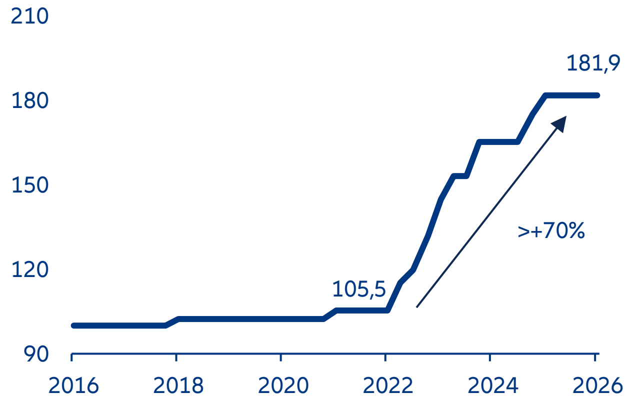


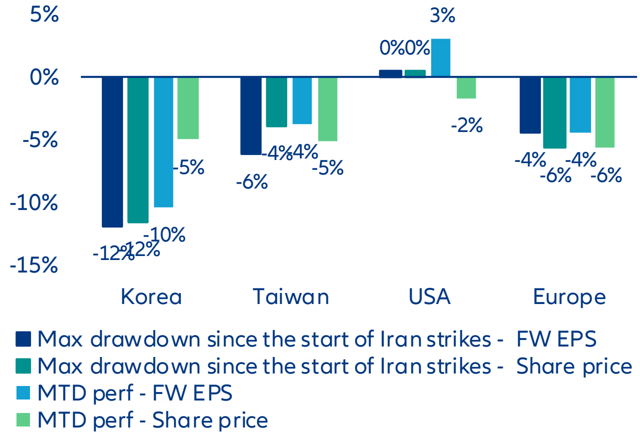
Data as of March 2026. Sources: LSEG Datastream, Allianz Research

The industrial pressure point is more immediate and more concrete. South Korea's industrial power price has surged over +70% since 2022, and both South Korea and Taiwan – home to the world's most critical semiconductor fabrication capacity – remain structurally dependent on Qatari LNG and helium supply as primary inputs for producing wafers. Any supply disruption or price spike flowing from Middle East instability hits Asian chip manufacturers directly in their cost base, compressing margins and risking output constraints at the worst possible moment for an AI ecosystem already running on tight GPU supply.

Is the capex cycle at risk from a chip/critical material shortage? At this stage we consider that risks of disruption in the super investment cycle are limited for two reasons: 1) capex is mostly driven by cash-rich and counter-cyclical companies whose revenue is less dependent on the macroeconomic outlook and 2) AI-related demand remains the top priority across the semiconductor supply chain. Regarding that second point, even moderate disruptions in wafer output are unlikely to materially derail ongoing capex dedicated to AI infrastructure. However, such constraints would limit any near-term acceleration in investment. They could also delay project timelines and increase implementation costs. However, a more volatile market environment will likely reinforce investor caution, particularly regarding datacenter monetization and the visibility of sustainable AI-driven revenue growth. The geopolitical stress does, however, carry one structural positive: it intensifies the urgency of supply-chain diversification, opening a credible investment case for semiconductor capacity in Europe and the US – a rebalancing that could, over time, reduce the systemic concentration risk that currently makes the entire AI buildout hostage to a handful of facilities in a narrow geographic corridor.

Figures 13 & 14 : Industrial power price in South Korea / EPS & share price variation of tech stocks since 27 Feb.

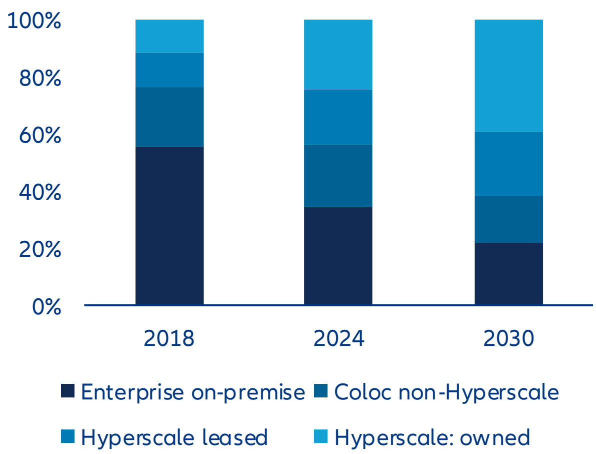




Data as of March 2026. Sources: Public statement from KEPCO, LSEG Datastream, Allianz Research

Smaller but smarter: monetization debate might lead further rationalization. Slowing AI adoption, compressing returns and volatile markets will inevitably test the conviction behind current capex programs. Yet reducing investment is not a credible option: pulling back signals retreat, cedes ground to competitors and risks losing the momentum that separates infrastructure leaders from fast followers. The capex expansion continues – but its risk profile is quietly being renegotiated. The scale of what is being built is extraordinary. Global data center capacity is on course to double by 2030, from roughly 130 GW today toward 260 GW, with hyperscalers as the overwhelmingly dominant engine – their share of total IT load expanding nearly 20pps to exceed 60%. At a 23% capacity CAGR, that implies approximately 100 GW of net new hyperscale deployment over five years. Applying current AI GPU cost structures and next-generation energy intensity assumptions, GPU procurement alone runs at roughly USD16bn per GW – implying a gross annual capex requirement approaching USD960bn if hyperscalers were to fund every megawatt themselves. But they won't bear that financial burden alone. Assuming current leasing ratios hold – where hyperscalers procure IT equipment directly but lease shell and core infrastructure – the annual capital commitment compresses to approximately USD730bn, a figure that aligns squarely with consensus estimates of USD760bn for 2028 alone. The next strategic frontier is pushing that ratio further: higher leasing penetration, more aggressive cost-sharing with co-investment partners and procurement structures that strengthen the AI service arsenal while keeping the balance sheet breathing.

Figures 15 & Table 4 : Distribution of current and future data center IT power load capacity per company profile breakdown / Simulation of yearly capex requirement from hyperscalers over the period 2026-2028 to fund datacenter estimated expansion



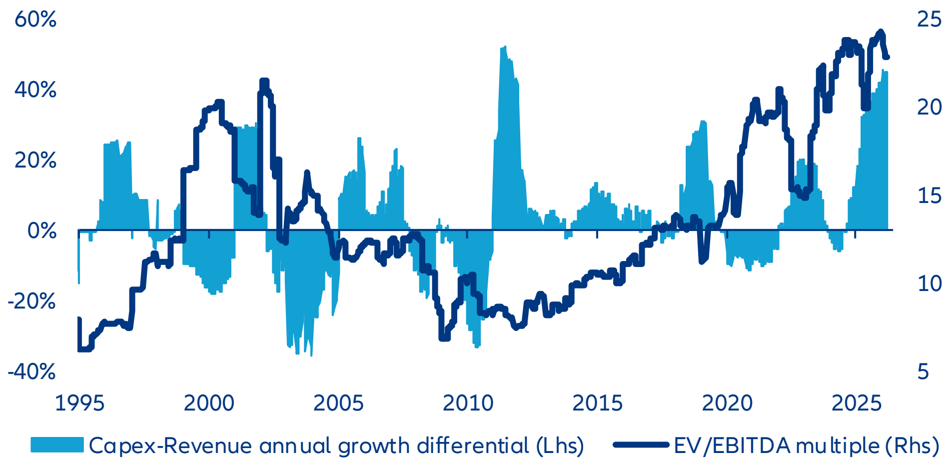
GPU ratio in total DC construction cost / DC full ownership ratio	40%	30%	Average
100%	827	1102	964
75%	703	909	806
63%	645	819	**732**
60%	628	793	711
50%	579	716	647
40%	529	639	584
30%	479	562	521

Sources: Synergy Group Research, Allianz Research

The changing economics of Big Tech: AI investment, profitability and valuation

Too highly valued? Capex monetization is the new compass for market mood. Despite the recent correction, US technology and AI equities remain richly valued relative to earnings, with EV/EBITDA multiples near 25x, close to historical extremes and above telecom valuations preceding the 2000 dot-com peak. The core issue is no longer AI potential but timing: capex is expanding far faster than revenues, with a ~46% growth gap between investment and sales, exceeding the 32% divergence observed during the 2001 telecom excess cycle. Infrastructure is scaling ahead of monetization, forcing investors to question whether current valuations already price in profits that remain distant. As early expectations of rapid AI earnings acceleration fade, markets are recalibrating toward execution risk and cash-flow delivery. This cooling of sentiment may ultimately mark a healthy transition – from narrative-driven valuation to discipline anchored in tangible returns.

Figure 16: Relative valuation and growth differential of revenue and investment from US technology industry

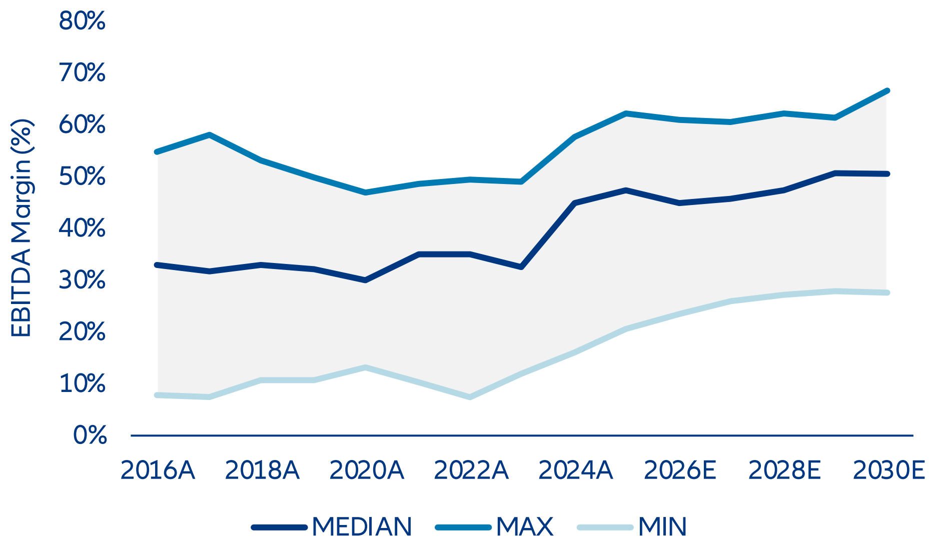


Data as of 16 march 2026. Sources: LSEG Datastream, Allianz Research

What are investors in Big Tech implying for AI growth and profitability? Assessing the profitability of AI-related activities remains challenging at the current stage of development. We are not trying to forecast the growth and profitability of AI companies but to form an opinion on legacy business contribution vs. AI expectations. We are also not trying to assess individual winners or losers but gauge sensitivities of the AI-complex to key assumptions. As a starting point we take consensus and past dynamics of legacy businesses. Segment disclosures and SEC filings (Forms 10-K and 10-Q) can be used to model legacy business activities separately, enabling a consistent projection of underlying operating performance. Given ongoing uncertainty surrounding AI monetization, particular emphasis is eventually placed on sensitivity analysis across key financial variables, with return on equity (ROE) serving as the central link between operating performance and valuation outcomes.

The changing capital intensity increasingly shapes the outlook for profitability. We focus on ROE which best combines the trade-off between capex and monetization. As illustrated in Figure 15, EBITDA margin development suggests that the expansion phase observed in previous years is gradually moderating. Amazon is an outlier at the lower bound, reflecting the still significant contribution of its lower-margin retail business, while hyperscalers remain influenced by the ongoing capex expansion cycle. Profitability remains structurally elevated, yet incremental improvements appear limited. This reflects the structural characteristics of large technology companies, many of which already operate with comparatively low cost bases and optimized operating structures. In fact, somewhat higher EBITDA margins are to be expected for a more capital-intense business and still imply lower net profitability. However, we can resonate cloud margins with the insights we have on existing cloud businesses, e.g. from cloud providers. The sensitivity analysis shown in Figure 18 highlights what we see as key drivers for financial performance and investor assessment via valuation. Variations in EBITDA margins exert a slightly lower influence on long-term outcomes, whereas changes in revenue growth assumptions more materially affect projected profitability as measured in ROE. This is also reasonable as transparency on data center profitability can already be better judged from existing operations than the unknown revenue potential from additional AI adoption and speed.

Figure 17: EBITDA margin development and peer range (2016–2030E)



Sources: Bloomberg Intelligence, Allianz Research

With the profitability engine under scrutiny, ROE sensitivity will gain attention on top of sales growth. Within our financial analysis, capex monetization via revenue growth therefore emerges as the primary focus channel for financial performance, which we gauge by ROE. At the same time, the analysis highlights a delicate balance between investment intensity and returns. Our capex-to-sales sensitivity further illustrates this trade-off, as higher investment intensity would reduce ROE across identical growth scenarios, with returns declining from around c.25% to c.23–24%, even before considering potential margin pressure from faster hardware depreciation. In this framework, a capex-to-revenue ratio of around 23%, as discussed in the previous analysis of AI investment dynamics, would broadly correspond to the +5% capex scenario in our sensitivity table and therefore reflects a moderate 0.4pp increase in investment intensity relative to the base case. At the same time, the previously discussed estimate of around 200bps margin compression from faster hardware depreciation would broadly be captured by the -2% EBIT margin scenario in the sensitivity analysis and would translate into roughly a 1pp reduction in ROE, broadly consistent with the sensitivity illustrated in our framework. This indicates that expanding the capital base alone does not enhance profitability unless supported by sufficient revenue monetization. Big Tech companies rush into capex to ensure market share in the structural growth trend. This comes at the risk of higher capital base with insufficient monetization. A lower capital expenditure ratio combined with stronger revenue realization support higher ROE outcomes. Big Tech companies therefore appear to operate along a narrow path between sustained investment and maintaining stable return expectations.

Table 4: ROE sensitivity to growth, margin and capex assumptions

ROE		EBIT-Margin Increase				
		-5%	-2%	0%	2%	5%
5Y Sales CAGR	-5%	21.81%	22.59%	23.11%	23.63%	24.41%
	-2%	22.10%	22.89%	23.42%	23.94%	24.73%
	0%	22.29%	23.09%	23.62%	24.15%	24.94%
	2%	22.49%	23.29%	23.82%	24.36%	25.16%
	5%	22.78%	23.59%	24.13%	24.67%	25.48%

ROE		5Y Sales CAGR				
		-10%	-5%	0%	5%	10%
CAPEX	-10%	24.35%	24.86%	25.36%	25.87%	26.37%
	-5%	23.92%	24.43%	24.93%	25.44%	25.94%
	0%	23.50%	24.00%	24.51%	25.01%	25.51%
	5%	23.07%	23.57%	24.08%	24.58%	25.09%
	10%	22.64%	23.14%	23.65%	24.15%	24.66%

Source: Allianz Research

ROE is key – beyond being a profitability indicator it is also one driver for fair valuation. Forward valuation metrics based on consensus estimates suggest broadly stable to slightly moderating P/E ratios from c. 35x this year to c. 23-24x in three years. Technology equities therefore remain premium-valued, yet forward multiples indicate limited scope for further expansion. Market expectations appear increasingly focused on realized earnings delivery rather than additional valuation rerating. In this environment, future equity performance is likely to depend more strongly on the successful monetization of AI-related investments and sustained top-line growth, while multiple expansion plays a diminishing role and rests on long-term growth. High ROE is a key determinant for investors to inform a "fair" valuation multiple because it directly drives future earnings growth from retained earnings. What actually matters is the spread between ROE and cost of equity. The latter is also influenced by AI reshaping companies' capex and sales via increased leverage and potentially higher revenue growth variability. Beta has increased from below 1.2 in 2016-24 to 1.5 in 2024-26. We apply our analysis on ROE to evaluate potential PE ranges for Big Tech and conclude that current valuations appear fair under current assumptions but with some sensitivity to sales and capex intensity. Via this multiplier, future financial performance will add an even higher impact for share price performance. The framework presented in Figure xx illustrates a sensitivity analysis, showing how different assumptions for five-year sales CAGR and capex intensity translate into varying long-term valuation outcomes, with the indicated PE centering around a c. 23x earnings multiple by 2030, consistent with a structurally strong but more mature profitability profile. Sensitivity on sales growth is highest and above implication from capex which anyhow is unlikely to drop notably soon while upside surprises are more likely and can be expected to dampen both, EPS and ROE via higher depreciation as well as PE.

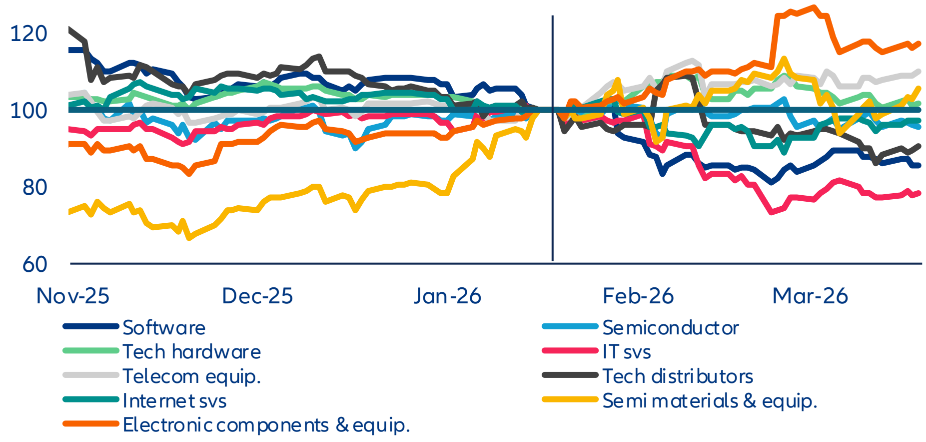
Table 5: Impact of growth and capital intensity on long-term valuation

P/E		Sales CAGR (surprise p.a.)						
		-9%	-6%	-3%	0%	3%	6%	9%
CAPEX (surprise p.a.)	-10%	23x	24x	25x	27x	28x	29x	31x
	-5%	22x	23x	24x	25x	26x	27x	29x
	0%	21x	21x	22x	23x	24x	25x	27x
	5%	20x	20x	21x	22x	23x	24x	25x
	10%	19x	19x	20x	21x	22x	22x	23x

Source: Allianz Research

The long-term outlook of the tech industry still looks rosy but uneven and inequal. We expect a more uneven performance trajectory for US technology stocks, with downside risks driven by valuation pressure and a broader shift toward risk-off positioning. Rotation away from megacap tech had already begun ahead of the Middle East escalation, suggesting that recent geopolitical events are amplifying, rather than initiating, the correction. Investor scrutiny is intensifying around the long-term monetization of elevated capex, particularly among hyperscalers, where the scale and timing of AI-related returns remain uncertain. At the same time, sentiment toward parts of the software sector is deteriorating, reflecting concerns over business model dilution as AI capabilities become more commoditized across corporate ecosystems. This combination is likely to sustain a degree of multiple compression and limit near-term upside for the most richly valued names. However, the outlook is not uniformly negative. Select segments within the tech value chain, notably semiconductors and infrastructure suppliers, continue to benefit from strong datacenter demand, with pricing dynamics potentially reinforced by rising commodity costs linked to geopolitical tensions. These areas appear structurally better positioned to capture near-term earnings visibility. In parallel, US leadership in AI infrastructure could attract incremental capital reallocation, particularly from Middle Eastern investors whose domestic investment pipelines may face delays. Over the longer term, the structural case for US tech remains intact, supported by innovation leadership and sustained demand for AI capabilities. That said, performance is likely to become more selective, with greater dispersion across subsectors and a less uniformly supportive environment than in previous cycles.

Figure 18: US equity performance of technology subsectors (rebased to 100 on 19 January)



Data as of 19 March 2026. Sources: LSEG Datastream, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends, (v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures, and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

The company assumes no obligation to update any information or forward-looking statement contained herein, save for any information required to be disclosed by law.

Allianz Trade is the trademark used to designate a range of services provided by Euler Hermes.

Allianz Research | 23 March 2026

Signal without response: Why the EU ETS needs resolve, not redesign

Patrick Hoffmann
ESG & AI Economist
patrick.hoffmann@allianz.com

Maximilian Bong-Maurer
Investment Strategist
maximilian.bong@allianz.com

In Summary

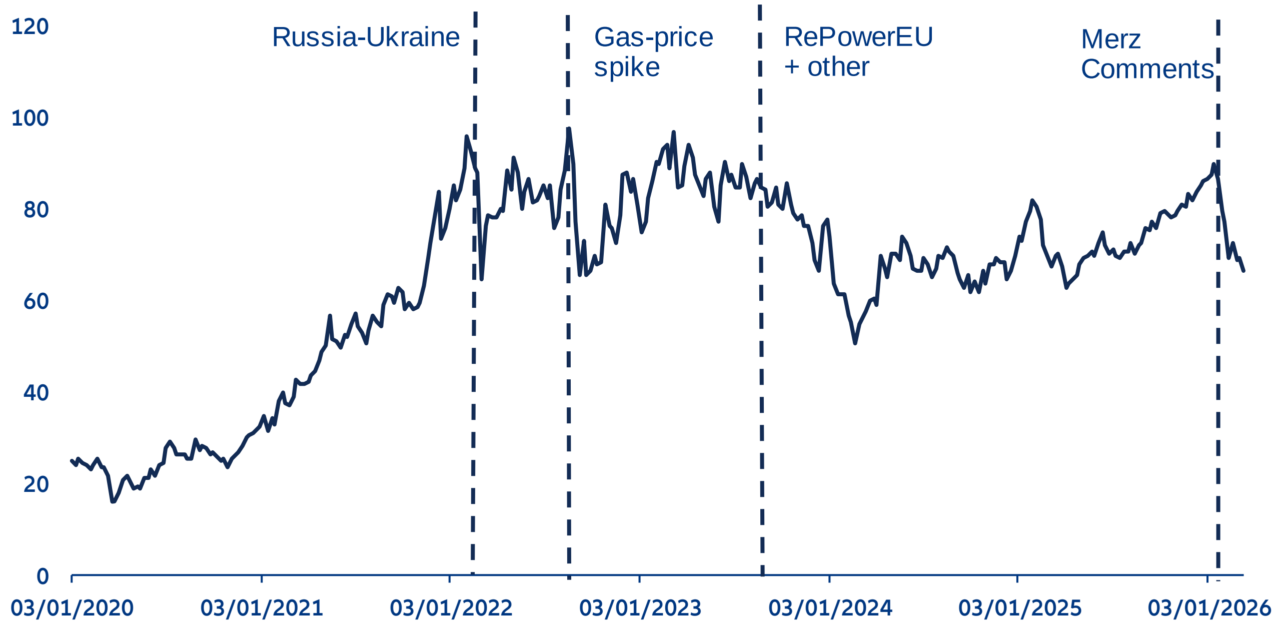
- **The EU Emissions Trading System (ETS) is at a political and structural inflection point, with allowance prices falling sharply by around -24% in the first two months of 2026. The conflict in Iran and associated energy price and inflation expectations have additionally spurred proposals to use the ETS as a lever for short-term price relief.** Unlike earlier episodes of price volatility rooted in energy-market fundamentals, the current turbulence reflects growing political questioning of the ETS's long-term design, compounded by concerns over industrial competitiveness as free-allowance phase-outs begin. Geopolitical pressure around Iran adds to this risk. The use of ETS revenues for price support or direct market intervention risk defeating the steering purpose of the system or creating dangerous lock-in effects.
- **Three ETS fault lines: free allocation has shielded more than 90% of industrial emissions from direct carbon costs but this is set to change; the ETS has driven substantial emission reductions in the power sector but not in industry and revenue recycling is falling short of its potential.** At current prices, full exposure would cost 0.9% of gross value added in the industry sector, a real burden for energy-intensive firms that cannot pass carbon costs through to customers given high energy costs, strong international competition and the limited offsetting potential of the Carbon Border Adjustment Mechanism (CBAM). Since 2005, power sector emissions have fallen by -54%, while industrial combustion emissions declined by a more moderate -33%. Renewables have made decarbonization viable at prices below EUR50/tCO2, while industrial options such as green hydrogen and low-carbon steel carry abatement costs well in excess of EUR100/tCO2. Persistent price uncertainty, with allowances oscillating in a EUR50–98/tCO2 corridor since 2022, has further weakened the investment case for capital-intensive industrial transformation. Last, only 16% of recycled revenues flow back into energy and industry, the sectors directly covered by the ETS, while 51% is directed toward non-covered sectors. A cumulative EUR41bn went undeployed for climate action between 2013 and 2024, risking a financing trap where industry bears rising carbon costs without the capital needed to decarbonize.
- **ETS2 is better positioned than ETS1 but that advantage is eroding.** Unlike ETS1, where abatement economics were the binding constraint, low-carbon alternatives in transport and buildings are already cost-competitive. The remaining barriers are political and financial: EUR113bn in annual fossil-fuel subsidies undermining the price signal, policy reversals on fossil boiler phase-outs, unaffordable EVs and financing frameworks still unequipped to meet household-level transition costs at scale. With household exposure potentially reaching EUR420 per year by 2030, these are policy choices that should be addressed before the carbon price arrives in 2028.
- **The ETS does not need a redesign but rather targeted reforms to restore price credibility and close the investment gap.** Seven priorities stand out: establishing a credible long-run EUA price corridor; scaling up Carbon Contracts for Difference to bridge the abatement cost gap in hard-to-abate sectors; introducing binding revenue recycling standards linked to the new decarbonization bank; prioritizing shared infrastructure for CO_2 transport, hydrogen and industrial clusters; conditioning the free allowance phase-out on the availability of abatement technologies and transition funding; expanding CBAM to cover downstream sectors currently left exposed and ensuring ETS2 revenues are transparently earmarked for households most exposed to its regressive cost distribution.

The ETS under fire: Market volatility and political pushback

The EU Emissions Trading System (ETS) entered 2026 under unusual strain. After reaching a two-year high of over EUR92/tCO2 in January, the EU Allowance (EUA) price corrected sharply by -14% within the same month and another -10% in February, falling to approximately EUR69/tCO2 (Figure 1). While the initial drop was triggered by geopolitical uncertainty over US tariffs, the future of Greenland and Iran, the second leg of the decline came amid growing political critique from member states, including Germany and Italy, which called into question not only the phase-out schedule of free allowances, but the long-term design of the mechanism itself. This turbulence arrives at a particularly consequential moment as the European Commission conducts its review of the ETS to determine whether its architecture remains fit for purpose in light of evolving market conditions and the EU's 2040 climate targets.

The observed price volatility is not unprecedented within the ETS, but earlier episodes were driven by fundamentally different forces. Between 2020 and 2022, allowance prices rose sharply, reflecting the post-Covid demand recovery, strengthened EU decarbonization ambition and an accelerated free-allowance phase-out schedule. The Russian invasion of Ukraine in February 2022 reversed this trajectory as anticipated gas supply disruptions curtailed industrial output and reduced compliance demand. Prices weakened further in 2023 amid declining industrial activity and the injection of additional allowances to finance RePowerEU. As Europe's energy dependence gradually shifted from Russian pipeline gas to US LNG, geopolitical risk remained the principal external driver of price formation, with US trade policy increasingly substituting pipeline politics as the key source of market uncertainty. Across these episodes, price movements were primarily rooted in energy market fundamentals or deliberate supply-side interventions, rather than in a broader questioning of the ETS's institutional design.

Figure 1: ETS allowance price and key market drivers (EUR/tCO2)



Sources: LSEG, Allianz Research

The key distinction from earlier episodes lies in the structural changes to the ETS taking effect from 2026, compounded by broader concerns about industrial electricity prices. From 2026 onwards, the free allowances allocated to European industry under the ETS begin to be phased out. The rationale was straightforward: as the Carbon Border Adjustment Mechanism (CBAM) enters into force to address international competitiveness concerns, the justification for shielding domestic industry through free allocation would progressively diminish, allowing the carbon price signal to drive further decarbonization. In practice, however, the transition has proved more contentious than anticipated. The CBAM, as finally adopted, covers a narrower set of sectors and a more limited range of emissions than originally proposed, leaving significant parts of European industry, particularly in downstream sectors, exposed to carbon costs without equivalent border protection. This concern is further reinforced by electricity prices that, while lower than their post-energy crisis peaks, remain structurally higher than those faced by key international competitors. At an EUA price of EUR65/tCO2, the carbon cost component of industrial electricity bills can reach up to 9% in the most carbon-intensive power systems, which disproportionately affects energy-intensive firms for which energy costs represent a larger fraction of total expenditure.[1]

Against this backdrop, political reactions have been swift. In Germany, the government announced plans to introduce a subsidized industrial electricity price of EUR0.05/kWh from 2026, roughly a third of the prevailing market rate, applicable to approximately 2,000 energy-intensive companies at an estimated fiscal cost of EUR3-5bn over three years. In Italy, a 2026 energy decree worth over EUR3bn proposed reimbursing gas-fired power plants for their ETS compliance costs, a measure that, given the role of gas in setting marginal electricity prices, would effectively decouple wholesale power prices from the carbon price signal. The European Commission has since opened a formal review of the measure's compatibility with EU state aid rules. Taken together, these interventions reflect a broader pattern in which the convergence of elevated energy costs and rising direct carbon-cost exposure is prompting member states to pursue unilateral measures that risk fragmenting the role of the ETS as a pan-European pricing instrument.

The conflict in Iran and associated energy price and inflation expectations have additionally spurred proposals to use the ETS as a lever for short-term price relief. These range from deploying ETS revenues for energy price support and adjusting the free allowance removal schedule to calls for direct market intervention to curb volatility. While some of these measures, such as the proposed ETS financed EUR30bn decarbonization booster, have their merits, others, such as the use of ETS revenues for price support or direct market intervention, risk defeating the steering purpose of the system or creating dangerous lock-in effects. The review of the ETS which was advanced to July 2026 is important and adjustments to the system can be warranted, however, the mechanism should not be misused as a short-term energy price management tool. For this purpose, other measures such as reducing non-steering levies on electricity including VAT, temporary state aid mechanisms for the most exposed energy-intensive industries, or targeted household transfers have a more direct and targeted impact on energy costs without undermining the carbon price signal the ETS is designed for.

Carbon costs in context: Limited exposure, rising trajectory

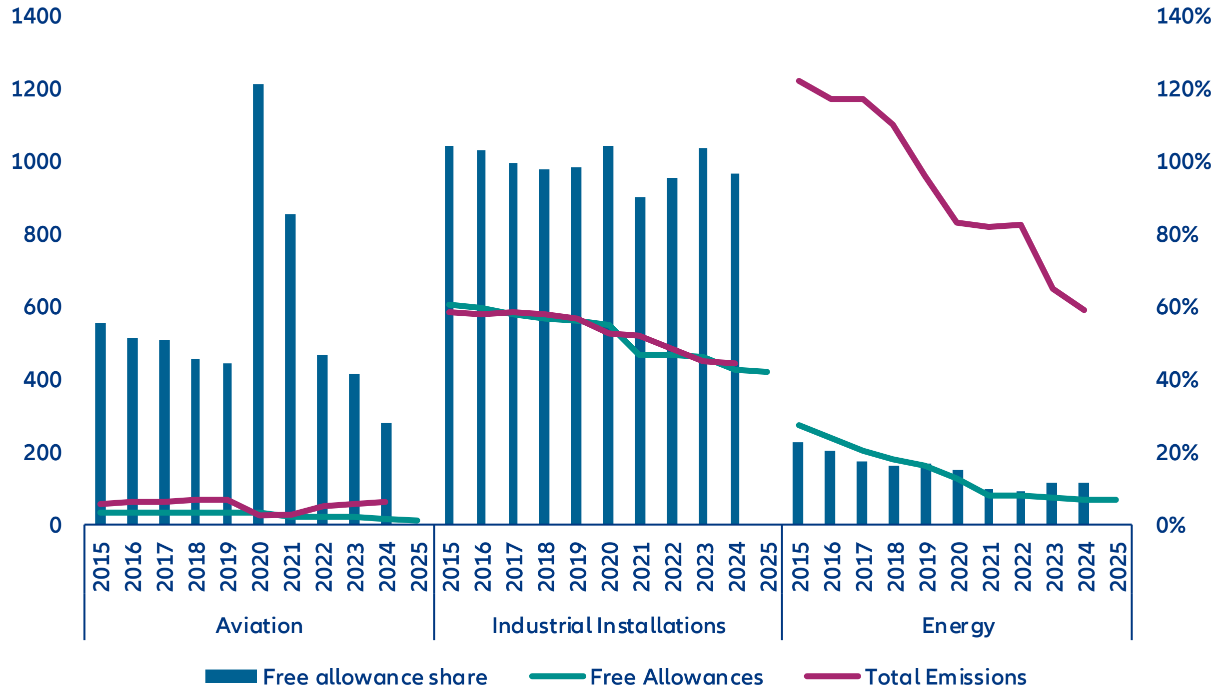
Despite the political concerns, the direct cost burden imposed by the ETS on European industry has thus far been limited. As shown in Figure 2, more than 90% of Scope 1 emissions in covered industrial sectors have been offset by free allowance allocation, effectively shielding most of the sector from direct carbon-cost exposure while some companies even generated profits from selling surplus allowances on the market. Even in a theoretical scenario with full emission coverage, ETS compliance costs in industry would amount to approximately EUR29bn, or around 0.9% of gross value added (GVA) across European industry in 2024 (Figure 3). The power sector bears a considerably higher burden in relative terms, at EUR38bn or 11.3% of sectoral GVA.

The concern is therefore less about the current level of exposure than its trajectory. Unlike utilities, industrial companies have limited ability to pass higher carbon costs through to consumers, making them more directly exposed to margin compression as free allowances are phased out. This is compounded by the gradual reduction in the overall supply of allowances, which will place upward pressure on EUA prices over time even

[1] ECB

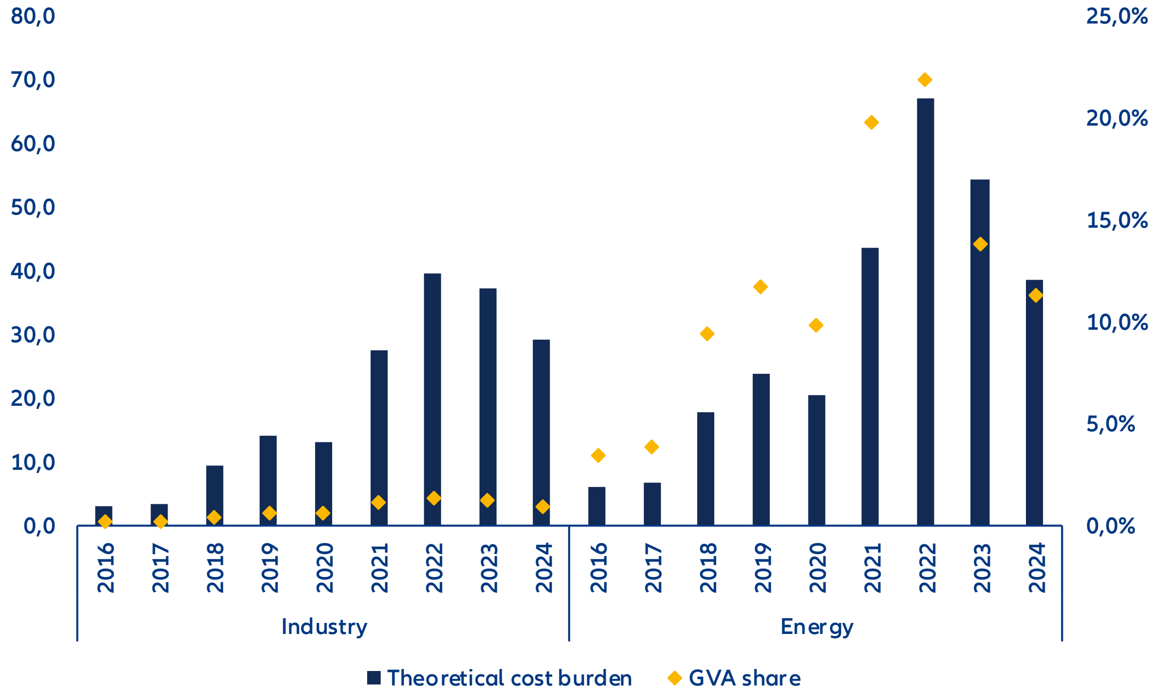
absent further policy changes. While the aggregate burden appears manageable at the economy-wide level, it is distributed unevenly across sectors and firm types, with energy-intensive industries facing the sharpest increase in net carbon costs as their allocation diminishes. A further structural issue is that the tradability of free allowances has not, in practice, generated the expected pace of industrial decarbonization. While the opportunity cost of holding rather than selling surplus allowances should in theory incentivize emission reductions, this mechanism has not translated into meaningful abatement investment across much of European industry. Understanding why requires looking beyond cost exposure to the sector-specific barriers that the carbon price signal has so far failed to overcome.

Figure 2: Sector emissions and free allowances (MtCo2, lhs) and free allowance share in total emissions (%, rhs)



Sources: EEA, Allianz Research

Figure 3: Theoretical cost burden under full exposure (EUR bn, lhs) and share in sector gross value-added (%, rhs)

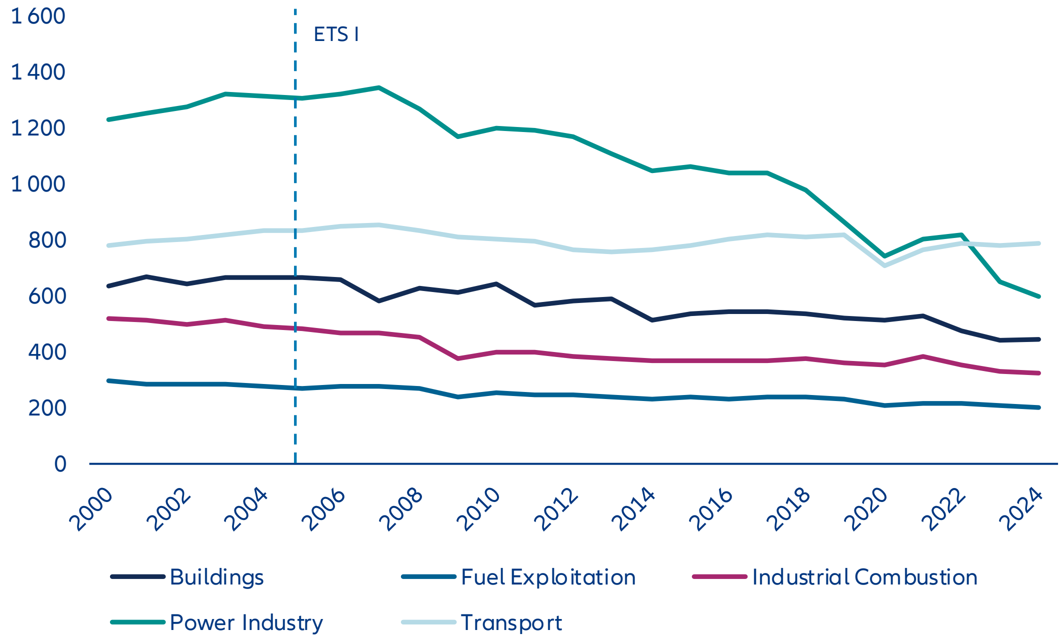


Sources: LSEG, Allianz Research

A tale of two sectors: Why carbon pricing works better for power than industry

The EU ETS has proven effective as a carbon-pricing instrument, yet its decarbonization impact varies markedly across sectors. Since its inception in 2005, emissions in covered sectors have fallen by more than -48%, compared with only -19% in non-covered sectors (Figure 4). However, the reduction is largely driven by the power sector, where emissions declined by -54%, while European industrial combustion emissions experienced a meaningful but more moderate decline of around -33%, reaching 324 MtCO2 in 2024. While free allocation is one mitigating factor in industrial decarbonization, firms retain incentives to abate as surplus allowances can be sold on carbon markets. The more fundamental constraints lie in sector-specific abatement economics. The power sector benefits from relatively short investment cycles, readily available low-carbon substitutes and competitive market structures that facilitate fuel switching. Industry, by contrast, faces longer capital stock turnover, technologically embedded process emissions and decarbonization options with higher marginal abatement costs – that is, higher costs for reducing an additional ton of CO2. These challenges are compounded by stronger exposure to international competition, which limits firms' ability to pass carbon costs through to product prices. As a result, carbon pricing may exert weaker investment and abatement signals in trade-exposed industrial sectors, even when allowances remain tradable.

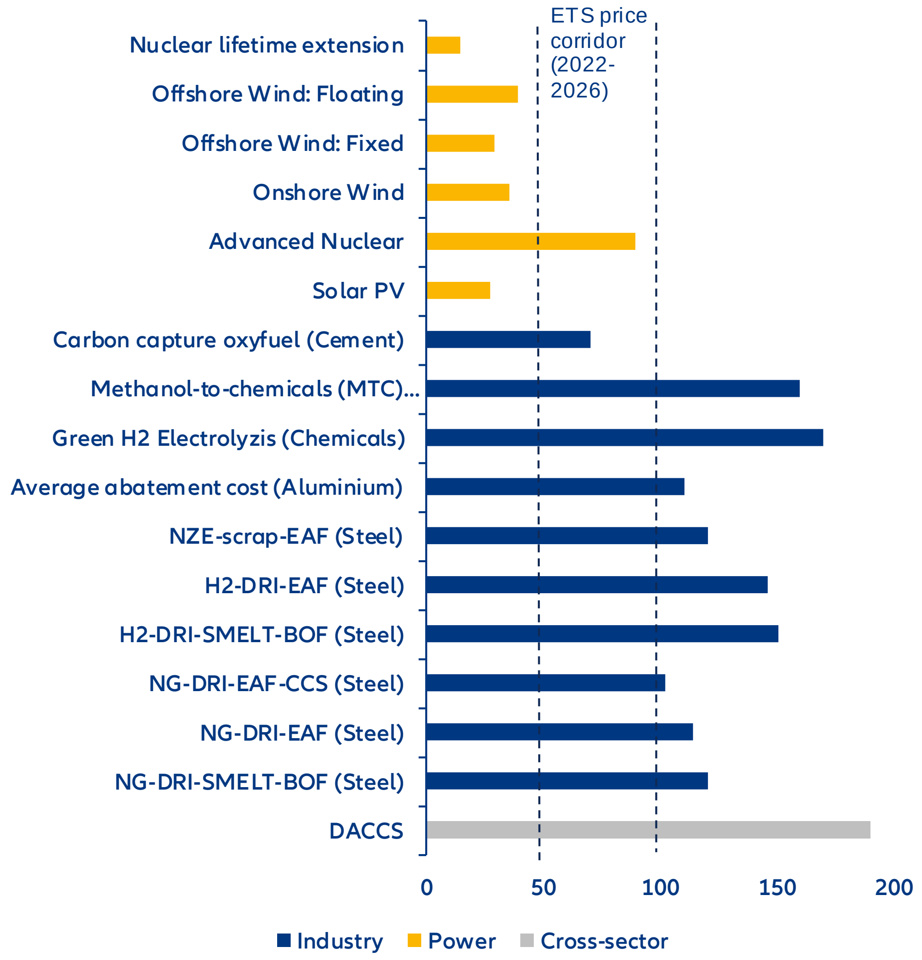
Figure 4: EU 27 emissions trajectory since introduction of EU ETS (MtCO2eq)



Sources: JRC EDGAR

High marginal abatement costs and price uncertainty represent the two principal barriers to industrial decarbonization under the current ETS framework. While the power sector has benefited from dramatic cost reductions in renewables that enable technology switching at prices below EUR50/tCO2, industrial decarbonization options such as green hydrogen, low-carbon steel and carbon capture typically carry abatement costs well in excess of EUR100/tCO2 (Figure 5). This cost gap reflects both the technological complexity of transforming energy-intensive production processes and the fact that many promising abatement technologies have yet to reach commercial maturity, leaving firms exposed to considerable cost and performance risk. Persistent price uncertainty adds a further barrier: although the post-Covid tightening of allowance supply did generate upward price pressure, since the energy crisis of 2022 allowance prices have largely oscillated within a EUR50–98/tCO2 corridor with considerable volatility. For capital-intensive investments with long payback periods and large upfront commitments, this unpredictability weakens the investment case precisely where long-term price credibility is most needed, and risks deferring the structural transitions that deep industrial decarbonization requires.

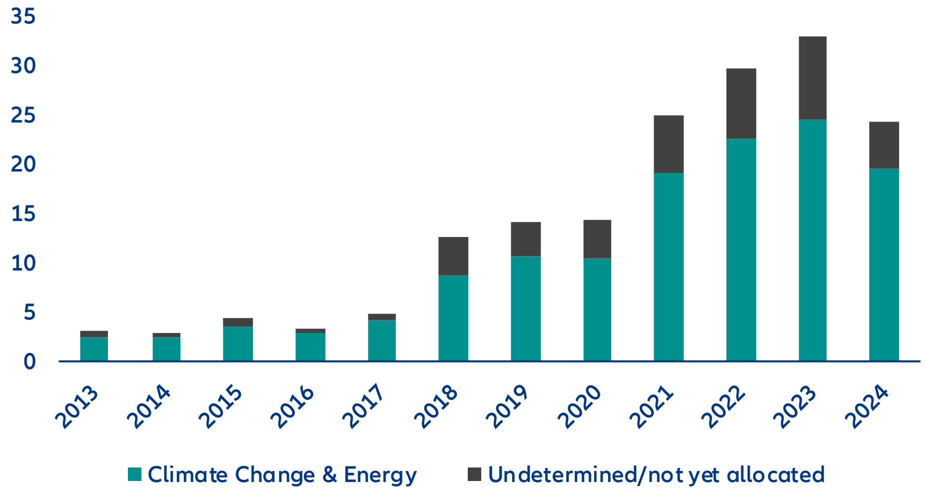
Figure 5: Abatement costs of selected decarbonization technologies by sector (EUR/tCO2)



Source: Allianz Research based on Agora & Wuppertal Institute [1, 2], MPP, EDF and IEA; Note: Abatement cost estimates can vary widely depending on sector, technology and region specific factors. Depicted values indicate cost averages.

ETS revenue recycling represents a significant underutilized lever for bridging the industrial-abatement cost gap. While the ETS obliges member states to reinvest allocated revenues in climate-related action, the mechanism underperforms in two key respects: the completeness of fund deployment and the strategic targeting of investments. Between 2013 and 2024, 13–31% of ETS revenues per year have remained unallocated or lacked a specific climate purpose, producing a cumulative deficit of approximately EUR41bn in effective revenue recycling (Figure 6). This structural shortfall limits the extent to which the ETS revenue mechanism can complement carbon price signals and support industrial transformation.

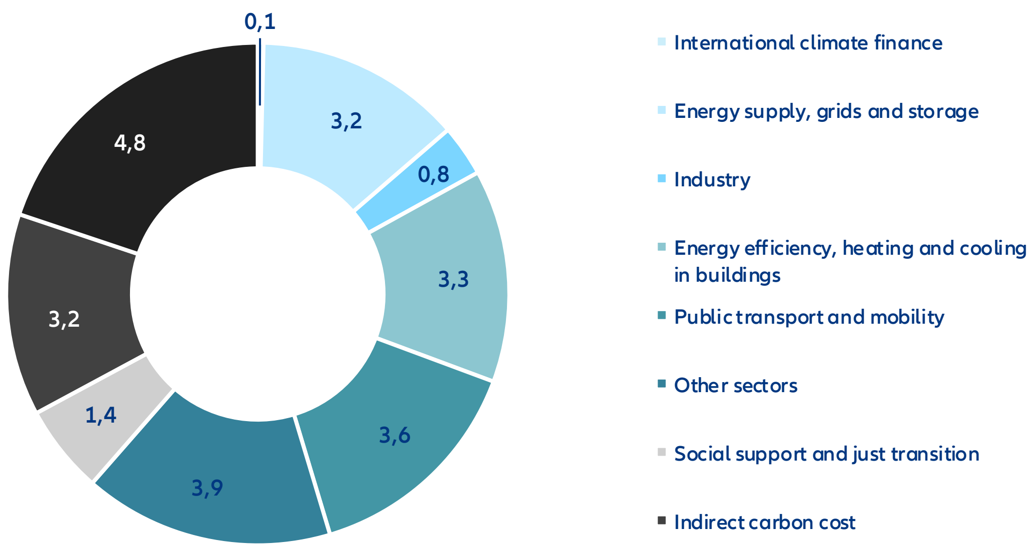
Figure 6: ETS revenue allocation gap (EUR bn)



Sources: EEA, Allianz Research

Of the funds that are recycled, the sectoral distribution raises further concerns. Only around 16% flows back into energy and industry transformation, the sectors directly covered under the ETS, while 44% is directed toward non-covered sectors such as buildings and transport (Figure 7). Although such cross-subsidization can be justified as a means of capturing near-term abatement opportunities and preparing these sectors for ETSII, an insufficient share returning to covered industries risks creating a financing trap, whereby firms bear the cost of carbon pricing without adequate access to the capital required to decarbonize. Equally important is the nature of recycling itself: in 2024, 13% of ETS revenues were directed toward "indirect carbon cost compensation" for energy-intensive industries, offsetting the very cost pressure the ETS is designed to create and thereby weakening the incentive to decarbonize. The same is true of support directed at technologies that have already reached cost-competitiveness, such as feed-in tariff schemes for mature renewables, where additionality is limited. To improve the efficiency of ETS revenue recycling, all revenues should be directed toward high-additionality decarbonization activities, ensuring covered sectors are not left underfinanced in the transition.

Figure 7: ETS revenue allocation by sector and purpose (%)

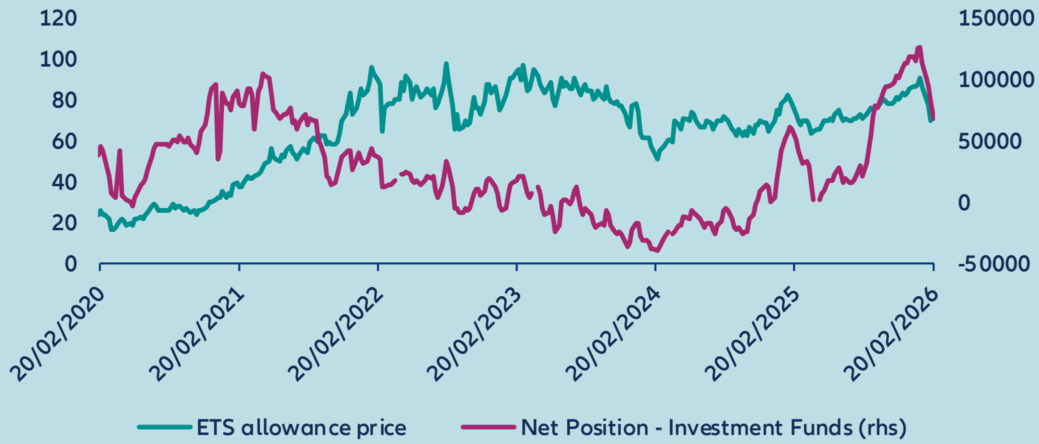


Sources: LSEG, Allianz Research

BOX 1: What is the effect on capital markets?

The financial sector constitutes a primary participant in EU ETS secondary-market trading by volume. While non-financial corporates and compliance entities accumulate net long positions, banks predominantly serve as counterparties on the short side. Investment funds represent a smaller but notable participant cohort, exhibiting positioning dynamics that more closely reflect prevailing market expectations for price developments. Following the Russian invasion of Ukraine, funds rotated to net short positions, consistent with anticipated contractions in industrial output and EUA demand (Figure 8). This positioning reversed to net long in November 2024, before funds began unwinding exposure in early January 2025, contributing to the pronounced price decline observed over that period. Prior to the price decline, investment funds had built up a record net-long position.

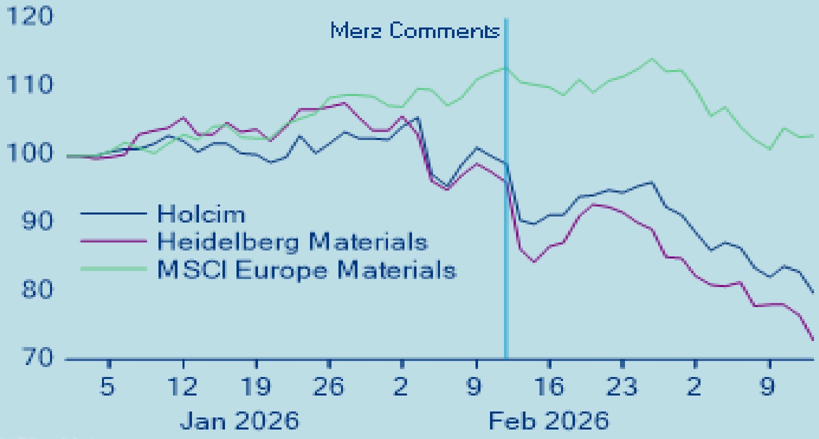
Figure 8: Investment funds' net positioning in EUA futures



Sources: Bloomberg, Allianz Research

The ongoing policy debate surrounding the future of the EU ETS has materially influenced investor expectations, particularly within the cement sector. Following the news of discussions on the future of the EU ETS market, the share price of Heidelberg Materials and Holcim, which are seen as transition leaders in the industry (see Transition Pathway Initiative assessment, for example), underperformed the broader materials sector (Figure 9). A prolonged phase-out of free allowances would disproportionately disadvantage companies that have already committed significant capital to decarbonization. Firms that invested early in low-carbon technologies or operational upgrades incurred substantial costs in anticipation of a tightening regulatory environment and rising carbon prices – an environment in which their abatement investments would generate a clear competitive return. An extension of free allocation effectively subsidizes higher-emitting peers, compressing the cost differential between decarbonization leaders and laggards and eroding the competitive advantage that early movers sought to secure. Simultaneously, lower prevailing CO_2 prices reduce the financial viability of capital-intensive abatement technologies such as Carbon Capture, Utilization and Storage (CCUS), for which the economic case depends critically on a sufficiently high and predictable carbon price signal. In aggregate, policy signals pointing toward a more lenient allocation regime risk undermining both the relative positioning of decarbonization leaders and the broader investment case for emission-reduction technologies across the sector.

Figure 9: Price reactions in European cement equities following statements on EU ETS reform



Sources: LSEG, Allianz Research

EUA price dynamics represent a meaningful driver of equity returns in carbon-intensive sectors. To quantify this relationship, sector index returns were regressed on both STOXX Europe 600 returns and the returns of rolled December EUA futures contracts from 2022 onwards. The results reveal a statistically significant and negative coefficient for EUA returns in the case of the STOXX Europe 600 Construction Materials Index and the STOXX Europe 600 Chemicals Index, indicating that rising carbon prices weigh on equity performance in these sectors. By contrast, no such relationship was identified for the STOXX Europe 600 Utilities or Basic Resources subindices. In the case of Basic Resources, this outcome is largely a function of index composition, given the limited share of steel producers among its constituents. For Utilities, the absence of a significant carbon price sensitivity is more structural in nature, reflecting the sector's substantial progress in decarbonization over recent years and the correspondingly reduced reliance on purchased allowances.

For now the market seems to price in an extreme scenario of adjustments to the EU ETS for some of the affected companies rather than a delayed phase out. Changes to the EU ETS are necessary to improve its effectiveness, but the phase-out of free allowances should continue. For investors aiming to decarbonize their portfolios and invest into transition leaders, these market reactions present a good opportunity to build up exposure to companies in hard-to-abate sectors that are actively decarbonizing their operations and that have credible transition plans.

Bracing for impact: Getting buildings and transport ready for ETS2

The EU ETS experience offers both a proof of concept and a cautionary tale for the next phase of European carbon pricing. Confirmed to begin in 2028 following a one-year delay, ETS2 will extend carbon pricing to fuel combustion in buildings and road transport, sectors that together account for 52% of EU greenhouse gas emissions and where decarbonization progress has significantly lagged. Unlike ETS1, ETS2 will operate as a separate emissions trading system with its own distinct carbon price, reflecting the different cost structures and abatement dynamics of these sectors. Regulated entities will be fuel distributors rather than end-users, who will bear the cost indirectly through fuel prices. To limit affordability pressures during the start-up phase, ETS2 includes a transitional price cap of EUR 45/tCO2 in 2023 real terms (approximately EUR59/tCO2 in nominal 2028 terms), below which the carbon price is prevented from rising freely. The central lesson from ETS1 is that carbon pricing works when abatement alternatives are cost-competitive and enabling infrastructure is in place before the price signal bites. The cap is therefore an acknowledgment that those conditions are not yet fully met, and that the transition needs time to mature before the full price signal can do its work.

Cost-competitive low-carbon alternatives exist in both sectors, but remain underdeployed. In road transport, battery costs have fallen by over -75% since 2015, yet European consumers have not fully benefited. Pack prices in Europe remain around 56% above Chinese levels, and carmakers' focus on high-margin premium models pushed the average EU EV price up by EUR5,000 between 2020 and 2024 to EUR44,600.[2] Passing battery cost reductions through to vehicle prices, alongside a continued expansion of charging infrastructure, will be essential to make EVs cost-competitive before ETS2 enters into force. In buildings, accelerating energy-efficiency renovations, expanding heating networks in urban areas and scaling heat pump adoption are the central levers, yet all three face high upfront costs that current financing frameworks are not yet equipped to meet at scale.[3]

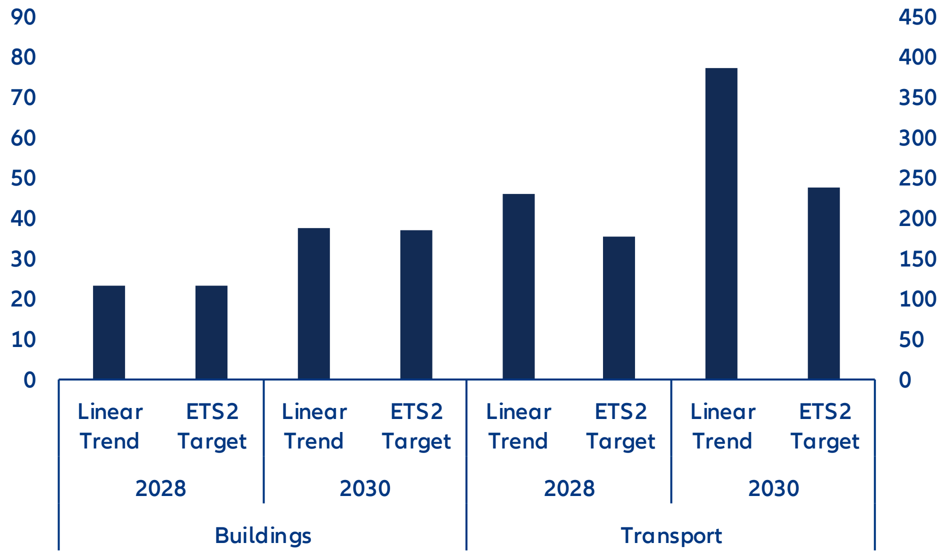
Closing these gaps before 2028 is urgent as inaction carries a measurable cost. In a pessimistic scenario with full sector coverage and only marginal reductions in emissions over the coming years, the annual ETS2 cost could be as high as EUR69bn by 2028 and EUR114bn by 2030 (Figure 10). Should ETS2 achieve its targeted 42% emission reduction against 2005 levels, these figures fall to EUR58bn in 2028 and EUR85bn in 2030. While this would alleviate some of the pressure on households, decreasing the cost burden by 15%-26%, the annual additional household expenses could come in as high as EUR420 per year by 2030. Reducing these costs is achievable, but hinges on adequate redistribution and well-designed supporting policies.

---

[2] *BNEF and T&E*

[3] *For a more detailed analysis on heat pumps see The heat is on: Unlocking Germany's heat-pump potential*

Figure 10: Projected ETS2 revenues per sector: annual total (EUR bn; lhs) and per household (EUR; rhs) under full sectoral coverage



Sources: JRC EDGAR, Eurostat, Allianz Research; Note: For 2028 an initial maximum allowance price of EUR59/tCO2 was assumed which increases to EUR100/tCO2 in 2030; The ETS Target scenario is based on the 42% ETS reduction target for 2030 while the linear trend assumes an annual emission reduction rate aligned with the average annual reduction over the last 20 years

Making ETS2 work requires addressing these conditions well before the carbon price arrives. ETS1 revenues have already been partially directed toward buildings and transport, providing a foundation on which ETS2 can build, but the gaps remain substantial. Fossil-fuel subsidies across the EU still amount to EUR113bn annually, suppressing the relative cost of carbon-intensive heating and driving and blunting the investment case for cleaner alternatives. A credible partial phase-out would simultaneously strengthen price incentives and free up fiscal space for the infrastructure and household support the transition requires. Equally important is policy consistency as repeated reversals on fossil boiler phase-outs, subsidy schemes and regulatory timelines have fostered a wait-and-see mentality among households and investors, deferring the long-cycle decisions on heat pump installations, building retrofits and EV purchases that need to be made within the coming years.

Social acceptability is the defining political challenge for ETS2, given that its costs fall more directly on households. The aggregate macroeconomic impact is manageable: at a carbon price of EUR 59/tCO2, the estimated inflationary effect amounts to approximately 0.54pp, while the GDP impact is projected at around 0.02pp provided revenues are recycled rather than absorbed into general fiscal accounts.[4] However, these aggregate figures mask a regressive distribution, with lower-income households facing a disproportionate burden as energy and transport represent a larger share of their expenditure. Ensuring that revenues are recycled toward those most exposed, through direct transfers, subsidized access to low-carbon technologies and investment in shared infrastructure, will therefore be central to maintaining the public legitimacy on which ETS2's long-term viability depends.

From price signal to investment signal: Seven priorities for ETS reform

The EU ETS is fundamentally sound, but the political commitment to making it work has not matched its ambition; what the system requires is not a redesign but the resolve to make it effective and predictable. The core challenge is not that the ETS is too stringent, but that it has failed to provide the certainty and complementary support that capital-intensive industries require to commit to deep decarbonization. That failure carries a forward-looking dimension: With ETS2 set to extend carbon pricing to buildings and road transport from 2028, the structural lessons of ETS1 – on price credibility, revenue recycling and enabling infrastructure –

---

[4] *See Climate policy: Europe's industrial shield or a drag on competitiveness?*

are also the design requirements that will determine whether the next phase succeeds. The following seven recommendations address this commitment gap and represent the priorities the Commission should act on in its 2026 ETS review:

1. **A credible and predictable EUA price path is a prerequisite for long-term investment planning in carbon-intensive industries.** Many of the most consequential decarbonization decisions, such as switching to hydrogen as an industrial feedstock, involve long capital cycles and require sufficient and stable carbon price expectations to be financially justified. The Market Stability Reserve has broadly fulfilled its role in correcting surplus-driven price weakness, but should be refined to better contain sharp volatility and support a credible long-run price corridor. Publishing a transparent indicative price trajectory would materially reduce investment uncertainty and strengthen the carbon price as a basis for industrial decision-making.
2. **Bridging the abatement cost gap requires scaling up Carbon Contracts for Difference (CCfDs) as a primary instrument for de-risking investment in breakthrough technologies.** CCfDs work by guaranteeing producers a fixed return for low-carbon production, paying out the difference between an agreed strike price and the prevailing carbon price when the latter falls short of what is needed to make the investment viable. This ensures that decarbonization investments in hard-to-abate sectors such as steel, cement, and chemicals remain financially attractive independent of carbon price volatility. Scaling up CCfD programmes and ensuring sufficient public backing to underwrite strike price commitments should therefore be a central priority of the ETS reform agenda.
3. **Revenue recycling must be made more effective through binding standards that prevent underspending and ensure funds are directed toward activities where additional decarbonization can be achieved.** The newly established decarbonization bank provides a potential vehicle for channeling industrial transformation funding at scale, and its mandate should be explicitly linked to ETS revenue flows. Alongside this, a coordination mechanism is needed to prevent unilateral national measures, such as subsidized industrial electricity prices or ETS cost reimbursements, from fragmenting the integrity of the pan-European carbon price signal. The same discipline must apply to ETS2 revenues from the outset: the EUR113bn in annual fossil-fuel subsidies across the EU that currently suppress the relative cost of carbon-intensive heating and transport represent both a fiscal and a political opportunity. A credible phase-out would simultaneously strengthen price incentives and free up resources for the household support and infrastructure investment that ETS2 requires.
4. **Decarbonization of energy-intensive industries also depends on shared infrastructure that no individual firm can finance alone.** Public investment should prioritize CO2 transport and storage networks, hydrogen pipelines and the development of industrial clusters where infrastructure costs can be pooled. Without this foundation, even well-funded firms face structural barriers to acting on the carbon price signal.
5. **The phase-out of free allowances should continue, but its pace must be made more explicitly contingent on the availability of abatement technologies, transition funding and infrastructure.** A phase-out that runs ahead of these enabling conditions risks creating a financing trap in which firms face rising compliance costs without a viable decarbonization pathway. The schedule should also incorporate a more systematic assessment of carbon leakage risk and competitive asymmetries, particularly for sectors not yet fully covered by CBAM.
6. **The Carbon Border Adjustment Mechanism should be expanded to cover downstream sectors and indirect emissions currently left exposed by its narrow adoption.** As free allocation is phased out, border protection must keep pace to ensure that European industry is not placed at a structural competitive disadvantage relative to international competitors operating under less stringent carbon regimes. A broader CBAM scope is therefore not only a competitiveness measure but a precondition for the phase-out of free allocation to proceed without undermining the industrial base the ETS is intended to transform.
7. **Social acceptability must be treated as a first-order design requirement for ETS2, not an afterthought.** Unlike ETS1, where carbon costs fall primarily on industrial firms, ETS2 will reach households directly through fuel prices for heating and road transport from 2028. The cost distribution

is regressive by nature – lower-income households spend a larger share of their income on energy and transport and will bear a disproportionate burden unless revenues are actively recycled toward direct transfers, subsidized access to low-carbon technologies and investment in shared infrastructure. Ensuring that ETS2 revenues are transparently earmarked for those most exposed is not merely an equity concern: it is the precondition for maintaining the public legitimacy on which the long-term viability of European carbon pricing depends.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed
or implied in such forward-looking statements.
Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends,
(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

The company assumes no obligation to update any information or forward-looking statement contained herein, save for any information required to be disclosed by law.

Allianz Trade is the trademark used to designate a range of services provided by Euler Hermes.

Allianz Trade

Allianz Research | 17 March 2026

Not all Emerging Markets are equal: Hormuz, triple deficits, and the new energy risk premium

Ludovic Subran
Chief Investment Officer &
Chief Economist
ludovic.subran@allianz.com

Ana Boata
Head of Economic Research
ana.boata@allianz-trade.com

Julia Belousova
Senior Emerging Markets Strategist
Julia.belousava@allianz.com

Lluís Dalmau Taulés
Economist for Africa and Middle East
lluis.dalmau@allianz-trade.com

Michael Heilmann
Senior Emerging Markets Strategist
michael.heilmann@allianz.com

Luca Moneta
Senior Economist for Emerging Markets
luca.moneta@allianz-trade.com

Giovanni Scarpato
Economist for CEE
giovanni.scarpato@allianz-trade.com

Garance Tallon
Economist for APAC
garance.tallon1@allianz-trade.com

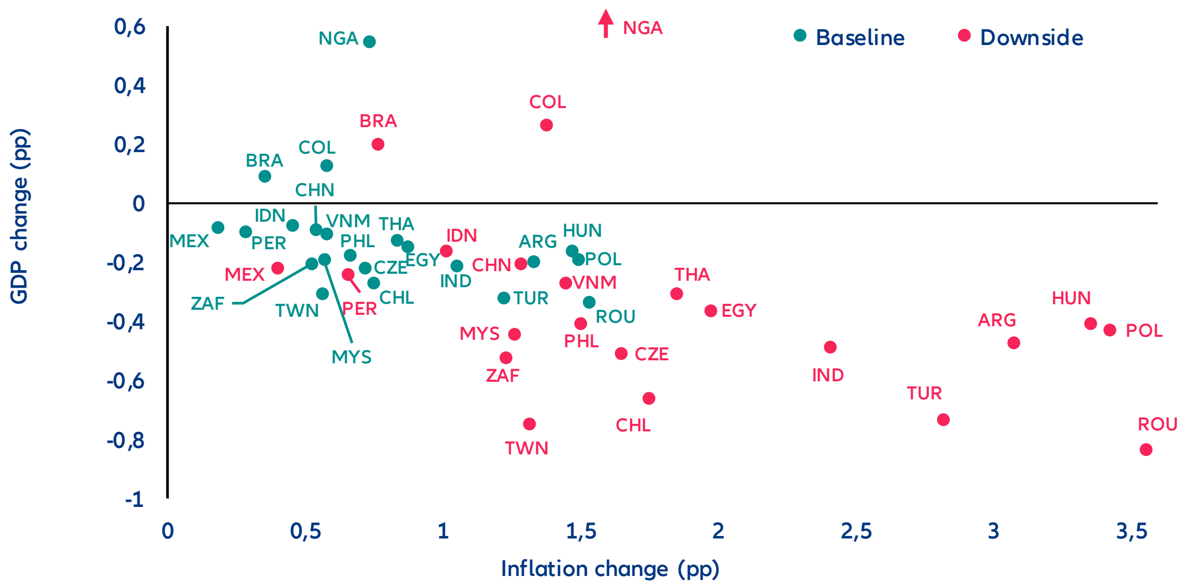
In Summary

- **Less than 2 months' disruption of the Strait of Hormuz could push average EM inflation higher by +0.8-1.0p with limited recessionary effects – apart from GCC countries.** We estimate that the closure of the Strait of Hormuz up to six weeks would result in a -1.6pps decrease in GDP for Saudi Arabia and -3.3pps for the UAE. Tourism – a key pillar of diversification across the Gulf – would also be hit in the short term, with knock-on effects on FDI and mega-project timelines including AI.
- **From price shock to supply shock? As the conflict drags on, Asian economies could face supply disruptions on top of stronger inflationary shocks, given that 56% of their oil imports and 30% of their total gas imports originate from the Middle East.** In Asia, Taiwan, Vietnam, Thailand, Pakistan, Bangladesh and Sri Lanka are more exposed to a supply shortage while in Africa, Egypt, Ethiopia, Kenya and Tunisia would suffer the most from a global oil supply shortage, given their exposure to Middle Eastern hydrocarbons. A temporary shortfall could be partially mitigated through adjustments in the energy mix (i.e. coal and renewables to a lesser extent), but energy supply disruptions for an extended period would require demand rationalization of -5 to -7% in the final energy consumption should prices double.
- **Beyond three months of closure for the Strait of Hormuz, many more EM countries are at high recession risk as they run triple deficits (fiscal, current account, energy).** GDP growth impact would average at least -0.5pp to 3.1% for emerging markets excluding China. Bangladesh, Egypt, Ethiopia, Jordan, Kenya, Morocco, Pakistan, Poland, Romania, Sri Lanka and Tunisia would be most at risk. Meanwhile, a second group of economies sits in moderately high risk as they have more room to maneuver to support their economies: Chile, China, Hungary, India, the Philippines, Taiwan, Thailand and Türkiye. By contrast, large commodity exporters such as Brazil and Mexico appear structurally resilient despite fiscal deficits as energy exports cushion the impact of higher prices.
- **The shock came at a supportive moment for EM carry. It calls for selectivity along a targeted energy risk premium.** Early market repricing has begun: FX markets reacted quickly, with the Egyptian pound experiencing the largest depreciation (-9.2%), followed by the Hungarian forint (-8%) and the Chilean peso (-4.9%). Repricing in local bond markets reveals a highly differentiated picture. In Mexico, the rise in nominal yields is largely explained by higher inflation expectations, while in Central and Eastern Europe – where the sell-off has been strongest – markets are pricing a larger share of risk and liquidity premia. The shock also complicates the policy outlook: With energy-driven inflation risks rising, many EM central banks are likely to remain on hold for longer despite slowing growth. A prolonged conflict could see inflation expectations repricing more forcefully across EM curves, steepening local yield curves and delaying monetary easing cycles. Overall, the adjustment is likely to remain selective rather than systemic. The most likely outcome is therefore the emergence of a targeted energy risk premium across vulnerable EM economies rather than a broad-based sell-off across the asset class.

From price shock to supply shock?

The 40% spike in energy prices in the first week of the conflict is the most rapid transmission channel to the global economy. On 9 March, the oil benchmark briefly reached USD120/bbl, later dropping to USD80/bbl. At the time of publication, oil remains around 15% above its pre-conflict level. Even if energy transportation is disrupted for only a short period, the conflict will still have a lasting impact on energy prices as it would take several weeks for production and supply to go back to pre-conflict levels. We expect prices to return closer to USD70/bbl, which would still be 16% above the pre-conflict baseline. This shift could push average EM inflation higher by 0.8-1.0pp, but the magnitude will depend on countries' energy exposure (particularly across Asia and energy-importing EMs). CEE economies – particularly Hungary and Romania – face the largest GDP hit given acute energy import dependency, followed by Thailand and Chile. Oil exporters such as Nigeria, Colombia, and Brazil would be relatively shielded, while Indonesia stands out among importers as more resilient given significant domestic energy production capacity.

Figure 1: Change to inflation and GDP growth rates in 2026 in selected EMs, baseline vs downside scenario



Sources: Oxford Economics, Allianz Research

The most immediate economic consequences are concentrated in the countries closest to the conflict — across the Gulf and the wider Middle East — where normal economic activity has been directly disrupted by the conflict. Early estimates of a short-term conflict show a GDP drop for Saudi Arabia and the UAE of -3pps and -4.3pps, respectively, considering major economic disruptions, the fall in tourism and a temporary drop of energy exports due to the effective closure of the Hormuz Strait. However, a short-lived conflict would also bring a quick rebound, with rates of +6.5% in Saudi Arabia and +7.6% in the UAE. The GDP loss in the entire Gulf Cooperation Council would be -3.3pps, with a considerable rebound of +6.4% of GDP growth in 2027. Dubai's real estate sector — one of the world's most dynamic property markets in recent years — took a significant hit on the Dubai financial market in the week since the conflict began, with main listed developers repricing sharply between -13% and -17%: The rest of the region, specially Kuwait and Bahrain, will be the most impacted given its dependence on the Hormuz Strait for both exports and imports. Bahrain's sovereign CDS spreads have widened by nearly 40% since the outbreak of hostilities — the sharpest move within the GCC complex and a clear market signal that investors are differentiating sharply between Gulf credits on the basis of fiscal resilience and hydrocarbon self-sufficiency.

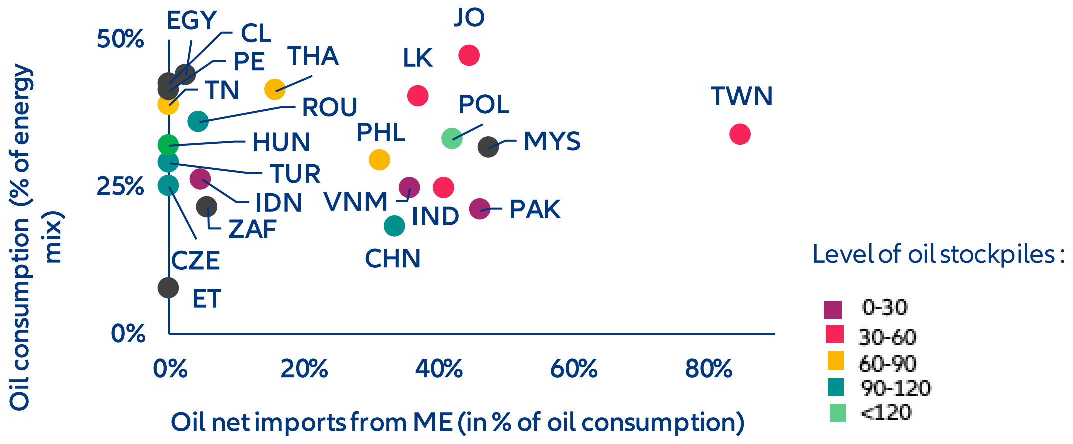
The long-term war scenario would result in harsher consequences via a large drop in hydrocarbon exports and reduced confidence in the region. History indicates several scenarios for how the conflict could impact the region. The Iraq-Iran tanker war in the 1980s, which threatened the closure of the Strait of Hormuz, had large long-term implications for Saudi Arabia, with the Kingdom growing at an average of +0.5% between 1981-1988 due to lower exports of oil. While both Saudi Arabia and the UAE both have some capacity to continue oil exports through different pipelines that connect their oil fields to ports outside of the Gulf, a sustained closure of the Hormuz Strait would greatly reduce both countries' hydrocarbon exports. Kuwait, Qatar, Bahrain and Iraq, which do not have much capacity to export without circulating the Strait of Hormuz, would suffer the most. Saudi Arabia and the UAE have both also enjoyed large inflows of investment and people, especially since the pandemic due to their open borders, tax exemptions for residents and increasing leisure and tourist attractions. Historical precedent suggests Gulf tourism could recover within one to two years. But if the conflict drags on and reshapes the region's security perception durably – as happened with Egypt between 2013 and 2017 – the recovery timeline extends to five to seven years, with compounding effects on FDI, mega-project timelines, the broader Vision 2030 ambitions and the region's ambition to become a global AI hub. The physical risk of data centers (Iran has targeted at least three data centers operated by Amazon in the UAE and Bahrain) and the consequences that this brings to economic activity in the country as it could paralyze banks, government offices, etc. Beyond the GCC, the rest of the region could be further impacted, with countries as far as Egypt or Türkiye recording drops in tourism.

A second wave of impact would come from hydrocarbon shortages from a longer closure of the Strait of Hormuz. This would turn into a significant supply shock – a qualitatively different and more damaging scenario that in most cases would lead to a recession in countries dependent on energy from the Gulf. Asia would be the most impacted region, given the high dependence on Middle Eastern hydrocarbons, with 56% of the region's oil imports coming from the Gulf. Gas prices could potentially double, which could trigger a fall in gas consumption of 5-7% in Asia. Building from the European precedent, we estimate that every -1% drop in gas consumption would equal 0.08–0.10pp GDP growth loss. Hence, for Asia, this could mean an impact above -0.5pp of GDP for the most impacted countries such as South Korea (-0.7pp) or a drop below 0.2pp as it is the case for India (-0.1pp).

The impact across Asian economies would be asymmetric, with Taiwan, Pakistan and Vietnam among the most impacted while Malaysia, Indonesia and China would remain shielded. Taiwan emerges as the most exposed country through its high dependence on Middle Eastern hydrocarbons, high dependence on oil in the energy mix and very low levels of oil stockpiles domestically. Pakistan and Vietnam are also in a delicate situation, with around 40% of oil consumption dependent on Middle East sources and very low reserves. However, their energy mix is slightly less dependent on oil than other EMs with a similar dependence on the Middle East but higher reserves. Despite reserves of around a month, India, and more particularly Sri Lanka, present similar characteristics and are also highly exposed. Thailand is another economy at risk: Despite relatively well-filled reserves and a low dependence on the Middle East, the country's high reliance on oil in its energy mix could result in a higher inflationary impact as it will need to buy more oil compared to peers. Similarly, if the conflict and closure of the Hormuz strait were to last, the Philippines and Indonesia could also start seeing some inflationary impact as more and more countries start competing for new partners. Despite a relative dependence on imports from the Gulf, China benefits from one of the highest levels of reserves in the region combined with a relatively low dependence on oil in its energy mix. And Indonesia should be shielded from an inflationary shock despite low reserves, given its low dependence on the Middle East, but also on the rest of the world.

Non-Asian emerging markets are comparatively better insulated from the Strait of Hormuz disruption. European emerging markets are relatively shielded by their low exposure to Middle Eastern crude – with the exception of Poland – relatively low oil dependency in the broader energy mix and solid strategic reserves. Despite higher reliance on oil, Latin America countries are amongst the most isolated from the shock due to a very low reliance on Middle East imports.

Figure 2: Oil dependency on the Middle East in selected EMs



Note: Oil dependency is computed as net imports of crude oil from Middle East divided by total oil consumption.

Sources: IEA, UN Comtrade, Allianz Research

The final macroeconomic impact would also depend on the scale of government support measures implemented to cushion households and firms. Policy responses could significantly mitigate the inflation shock, as seen during the 2022 energy crisis in Europe. At that time, European governments deployed energy-support measures equivalent to roughly 4% of GDP over a year, including price caps, tax cuts and transfers. Assuming the shock remains roughly half the size of the 2022 European energy shock, Asian economies, given their higher dependence on imported energy, would likely require subsidies of around 0.25% of GDP (or about 2% annualized) to smooth the impact on households and firms. In the CEE region, Hungary has already introduced fuel price caps, increasing pressure on the fiscal deficit running already at -5.1%. In Asia, Indonesia, South Korea, and Japan are providing continued support to their populations through energy subsidies and oil price caps. In the Middle East, Egypt sits in a precarious position, subsidies remain substantial, with the 2025/26 budget already allocation 0.6% of GDP, with a fiscal deficit already at -10% and a matched external deficit, Cairo could fall into fiscal slippages. During the last shock, UAE and Qatar supported Egypt with large transfers of capital, but with new priorities in the Gulf capital's due to the war, it remains a question how Egypt could sustained a long-period of high energy prices.

Despite potential mitigations channels, energy supply disruptions would require demand rationalization. While part of the shortfall could be mitigated through adjustments in the energy mix – for instance via a temporary ramp-up in coal generation – and roughly half of the disrupted volumes could potentially be rerouted or offset by higher oil production elsewhere (notably in the US or Russia), supply substitution would only partially compensate for the disruption. In that scenario, energy-demand rationalization would become unavoidable, affecting both companies and households. Several Asian economies — from Japan to Thailand — are feeling the strain of the Hormuz closure and have moved quickly to manage supply, implementing measures ranging from fuel export bans and energy-saving policies such as four-day workweeks and remote working mandates, to ramping up oil purchases from Russia. The European experience in 2022 illustrates the scale of adjustment that may be required. At the peak of the crisis when gas prices tripled to 330 EUR/MWh, and for a dependency to Russia of more than 40% of total EU gas imports, Europe managed to reduce gas consumption by almost -20% driven by high prices, supply shortages and emergency policy measures.

The Middle East represents 30% of total natural gas imports to Asia, again with some disparities across individual countries. With 80% of its imports coming from the Gulf, India is the most exposed to the repercussions of the conflict on gas prices, followed by Vietnam and Indonesia. However, as natural gas does not represent a high share of these countries' energy supply, the impact should remain relatively muted. Though the shock will be much less than that of oil, Thailand, Taiwan and Pakistan should be carefully monitored as they present a larger share of natural gas in their consumption mix compared to the region combined with a relative dependence on imports form the Middle East, especially for Taiwan and Pakistan where the Gulf represents more than 25% of their consumption..

More severe and systemic spillover effects would materialize if the Strait of Hormuz remains closed beyond the short-term threshold, transforming what began as a price shock into a structural disruption. Each additional week of Hormuz closure compounds recessionary pressure as reserves deplete and supply shortages broaden. In a scenario extending beyond three months, EM GDP takes an average hit of -0.5pp or more, with the most energy-dependent economies bearing a disproportionate share. In that scenario, emerging markets running twin deficits – fiscal and current account – would face acute pressure as deteriorating energy import bills compound already-stretched external balances and financing needs. Those carrying a triple deficit – where an energy balance deficit layers on top – would be most severely impacted, facing simultaneous pressure on their currencies, sovereign spreads and growth trajectories, with limited policy space to respond.

Countries combining fiscal deficits, external imbalances and energy import dependence emerge as the most exposed to a prolonged disruption in Middle Eastern energy supply. Economies such as Romania, Poland and Tunisia display the clearest triple-deficit configuration, where sizeable fiscal deficits coincide with current account gaps and structurally negative energy balances. In these cases, higher oil prices would simultaneously widen external deficits, worsen fiscal dynamics through energy subsidies or weaker growth and weaken currencies through deteriorating terms of trade. Several frontier and lower-income importers – notably Egypt, Sri Lanka, Kenya, Ethiopia, Jordan, Morocco and Bangladesh – also appear particularly vulnerable given already elevated financing needs and limited macroeconomic buffers. In contrast, commodity exporters such as Nigeria, Colombia and Indonesia remain comparatively insulated as positive energy balances provide a natural hedge against higher oil prices, supporting both fiscal revenues and external accounts.

A second group of economies sits in an intermediate zone where vulnerabilities stem primarily from energy dependence rather than broad macroeconomic imbalances. Chile, China, India, the Philippines, Hungary, Taiwan, Thailand and Türkiye fall into this category. These economies run structurally negative energy balances and would therefore face pressure on external accounts if prices remain elevated, but the overall macro impact depends on policy space and external buffers. In several cases, relatively manageable current account positions, diversified economies or stronger reserve coverage could mitigate the shock. By contrast, large commodity exporters such as Brazil and Mexico appear structurally resilient despite fiscal deficits, as energy exports cushion the impact of higher prices. Overall, the table highlights that the transmission of a prolonged oil shock across emerging markets would remain highly uneven, with risks concentrated in economies where external imbalances, fiscal constraints and energy dependence overlap.

Table 1. Emerging markets exposure, market reactions and coping mechanisms

	Exposure to the Middle East crisis				Market reaction since Feb 27		Coping mechanisms					
Country	Energy balance (% of GDP)	Strategic oil reserves (days of consumption)	Vulnerability to energy price shock	Vulnerability to ME supply shock	FX (% change vs USD)	Bond yields (bps)	Fiscal balance (% of GDP)	Current Account (% of GDP)	FX reserves (months of imports)	FX outlook	Central Bank stance (by mid-	Risk of fiscal slippage throughout
Argentina	0.6	NA	Medium	Low	-0.04	11.0	-1.5	-0.4	3.8	Neutral	On hold	Low
Bangladesh	-3.5	49	High	High	-0.41	32.1	-3.9	-0.9	4.9	Negative	On hold	High
Brazil	0.9	NA	Low	Low	-3.77	88.3	-7.5	-2.3	12.1	Positive	On hold	Medium
Chile	-4.4	25	High	Medium	-4.94	18.5	-0.6	-2.2	6.7	Negative	On hold	Low
China	-2.4	100	Medium	Medium	-0.56	38.0	-5.0	2.8	12.3	Neutral	On hold	Medium
Colombia	3.9	NA	Low	Low	1.31	N/A	-5.7	-2.6	9.6	Negative	Hike	Medium
Czechia	-2.4	117	High	Medium	-4.47	58.8	-2.1	0.4	9.3	Negative	On hold	Medium
Egypt	0.0	NA	Medium	High	-9.16	40.0	-10.7	-4.3	4.3	Negative	On hold	High
Ethiopia	-4.0	NA	High	High	-0.47	N/A	-1.7	-2.6	3.7	Negative	On hold	Medium
Hungary	-3.6	215	High	Medium	-8.02	94.0	-5.1	0.9	3.7	Negative	On hold	High
India	-3.7	30	High	Medium	-1.61	2.0	-1.8	-1.4	9.1	Neutral	On hold	Medium
Indonesia	1.1	20	Low	Medium	-1.04	40.7	-2.7	-1.2	7.1	Negative	On hold	High
Jordan	-8.0	45	High	High	0	-50.7	-3.5	-5.5	7.7	Negative	On hold	Medium
Kenya	-4.5	NA	High	High	-0.27	10.0	-5.6	-3.4	5.7	Negative	On hold	High
Malaysia	-0.4	NA	Medium	Medium	-1.23	7.6	-3.6	1.8	5.3	Negative	On hold	High
Mexico	0.7	NA	Low	Low	-4.25	48.5	-4.1	-0.3	4.3	Neutral	On hold	Medium
Morocco	-5.0	30	High	Medium	-3.28	21.5	-4.5	-1.5	7.4	Neutral	On hold	Medium
Nigeria	12.2	NA	Low	Low	-1.91	61.3	-3.7	3.6	12.0	Neutral	On hold	Medium
Pakistan	0.0	28	High	High	0.02	66.0	-4.1	-0.4	11.6	Negative	On hold	High
Peru	-0.5	NA	High	Low	-3.1	30.7	-2.2	1.2	14.8	Negative	On hold	Low
Philippines	-4.2	60	High	Medium	-3.59	68.8	-0.3	-3.5	8.2	Negative	On hold	Medium
Poland	-2.1	121	High	Medium	-4.86	83.1	-6.3	-0.8	5.7	Negative	On hold	High
Romania	-1.7	92	High	Medium	-3.54	101.0	-6.2	-6.6	6.0	Neutral	On hold	High
South Africa	-2.3	NA	Medium	Low	-6.39	96.0	-5.6	-1.2	5.6	Neutral	On hold	Low
Sri Lanka	-4.5	35	Medium	High	-0.63	4.4	-6.5	0.5	4.5	Negative	On hold	High
Taiwan	-5.1	40	Medium	High	-2.25	4.0	-0.2	13.1	15.9	Negative	On hold	Medium
Thailand	-7.8	60	High	High	-4.58	33.5	-2.5	1.3	8.9	Negative	On hold	Medium
Tunisia	-4.0	60	High	High	-3.07	N/A	-6.0	-2.0	4.9	Negative	On hold	High
Türkiye	-3.6	94	High	Medium	-0.64	233	-0.7	-1.3	2.7	Neutral	Hike	Medium
Vietnam	-2.5	15	Medium	High	-0.89	7.7	-2.3	2.4	2.3	Negative	On hold	Medium

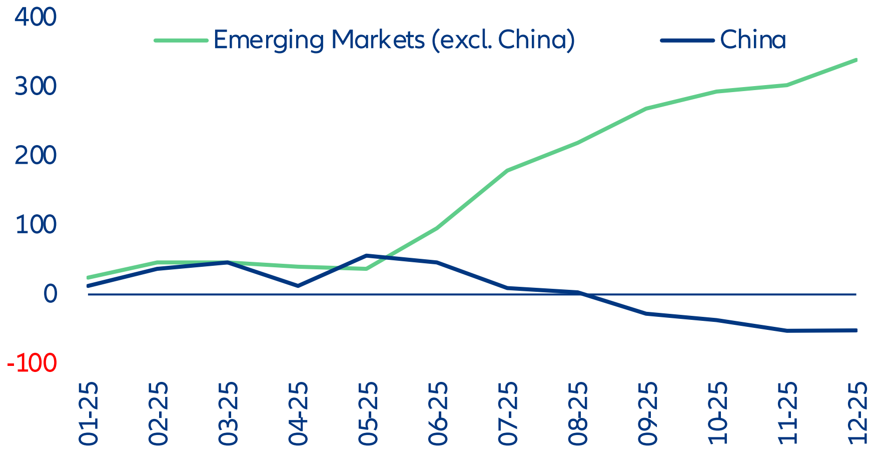
Sources: LSEG, Allianz Research. Note: Data as of 13/03/2026. In green central banks that were cutting before the conflict in the Middle East escalated.

Türkiye and India in particular sit at the intersection of competing blocs – geographically indispensable, politically non-aligned, economically interdependent – and risk paying a price. From a market perspective, both countries continue to post twin deficits (fiscal and current account balance). Last year, Türkiye returned to a positive primary balance for the first time since 2022, while India has had positive primary balance for some time. In the case of Türkiye, recent history shows that when energy prices were at their highest (2022, but also in the 2012-2014 cycle), Ankara has maintained a positive primary balance – at the expense of inflation. The current account will tend to be more negative this year, potentially hitting 2-2.5% of GDP if energy prices remain elevated, with effects on the lira and domestic inflation, but there may be room for subsidies. Fiscal discipline remains in place, but with debt/GDP at 25%, the value of gold reserves increasing further and the ever-present debate about bringing forward the presidential elections before their natural schedule in May 2028, the temptation to mitigate the effects of price increases in some way could also increase. India is also positioned with better footing than the 2022 scenario, with the fiscal deficit improved from the 2022 period (-0.8% in 2025 vs -2% in 2022) and the external balance at -0.2% of GDP. While an energy shortage could harm the country, India could shift oil and natural gas imports to domestic coal consumption, mitigating part of the shock. FX reserves remain very high at 11 months of imports.

Energy price premium ahead for emerging markets?

The new escalation in the Middle East could reverse the record gains of emerging markets. 2025 began with pronounced outflows for emerging markets (excluding China), reaching a cumulative low of -USD10.8bn around the implementation of President Trump's "Liberation Day" tariffs. But that marked a turning point as market sentiment and dollar depreciation turned in EMs' favor, bringing back flows. As a result, 2025 ended with record portfolio flows (USD410bn), almost double from the previous year, much above the historical average and a turnaround from the capital outflows experienced after the pandemic. EM bond funds concluded 2025 with their first annual net inflows since 2021, totaling +USD31.8bn.

Figure 3: Capital flows towards emerging markets, USDbn



Sources: LSEG Workspace, Allianz Research

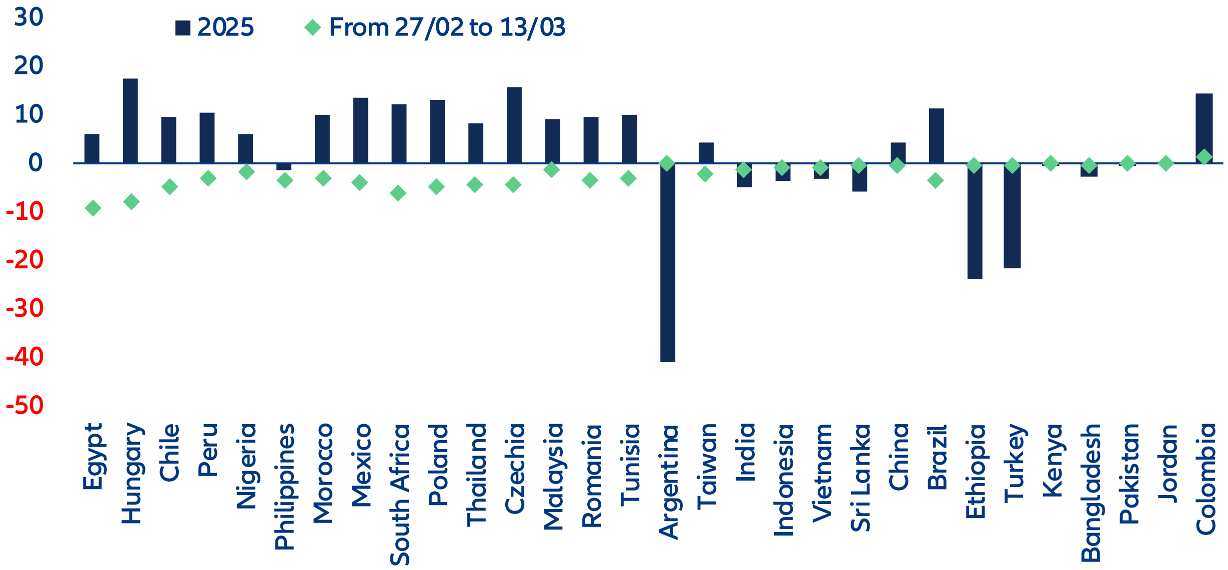
Looking to 2026, while the EM asset class is structurally stronger, selectivity will be an essential aspect of the path ahead. Direct benchmark exposure to the most vulnerable countries (Egypt, India, the Philippines, Thailand, Türkiye) remains limited, as they represent less than 3% of the MSCI Emerging Markets ex-China index. The most vulnerable European economies (Poland, Romania) would indeed feel the pinch of higher inflation but the risk of supply shortages would probably be offset by European mechanisms likely to be reinstated should the conflict last for longer.

Foreign exchange markets reacted quickly to the escalation in the Middle East conflict, with most emerging market currencies weakening over the week. Between 27 February and 13 March, several currencies recorded sharp declines as higher oil prices and a stronger US dollar triggered a broad risk-off move (Figure 4). The Egyptian pound saw the largest depreciation (-9.2%), reflecting its vulnerability as a large net energy importer with significant fiscal and external deficits. Central European currencies also weakened notably, with the Hungarian forint (-8%), Polish zloty (-4.9%) and Czech koruna (-4.5%) under pressure as the region's strong dependence on imported energy combined with the unwinding of investor positions. In Latin America, the Chilean peso (-4.9%) stood out among the underperformers given its negative energy balance, while in Asia the Philippine peso (-3.6%) and Thai baht (-4.6%) also weakened in line with the region's heavy reliance on Middle Eastern oil supplies. Part of the sell-off reflects the unwinding of earlier investor positioning: several currencies that had performed strongly earlier in the year were among the largest underperformers during the recent risk-off move.

As market conditions stabilize, currency movements are likely to more clearly reflect differences in countries' energy exposure. Higher oil prices tend to support currencies of energy exporters while weighing on importers, suggesting that the currencies of some Latin American exporters, such as the Brazilian real and Colombian peso, could stabilize once the initial risk-off adjustment fades. By contrast, currencies such as the Hungarian forint, Korean won, South African rand and Chilean peso appear to have already incorporated part of the energy shock and may remain sensitive to how the conflict and oil prices evolve.

Some currencies have remained relatively resilient despite their exposure to higher oil prices, notably the Indian rupee and Turkish lira, where active central bank management has helped limit volatility. However, the longer energy prices remain elevated, the harder it may become to maintain this stability. Looking ahead, the duration of the disruption in the Strait of Hormuz will remain the key factor determining further FX pressure: a prolonged disruption would keep energy prices elevated and sustain US dollar strength, while a faster resolution would likely allow part of the recent currency weakness to unwind.

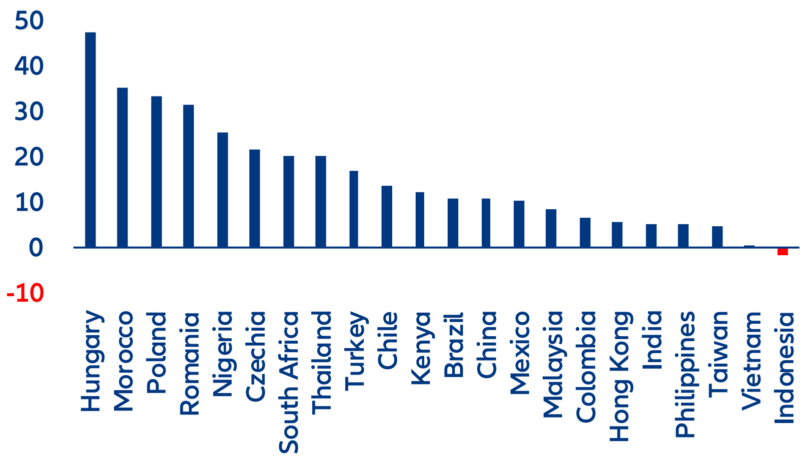
Figure 4: FX moves since the start of the Middle East conflict



Sources: LSEG Refinitiv, Allianz Research

Even with the ongoing shock, we continue to see EM fundamentals as strong, partially mitigating part of the war impact. FX reserves have remained elevated as many EMs used 2025 conditions to rebuild them, despite the challenges of US tariffs. India, South Korea, Taiwan and several Southeast Asian central banks recouped about USD132bn in FX reserves in late 2025 and early 2026, more than half of what they lost during earlier defensive currency interventions, aided by a weaker dollar and inflows. The Indian central bank has been particularly aggressive in rebuilding stockpiles to better defend the rupee on dips.

Figure 5: FX reserves have grown across most EMs over the last year (y/y growth)

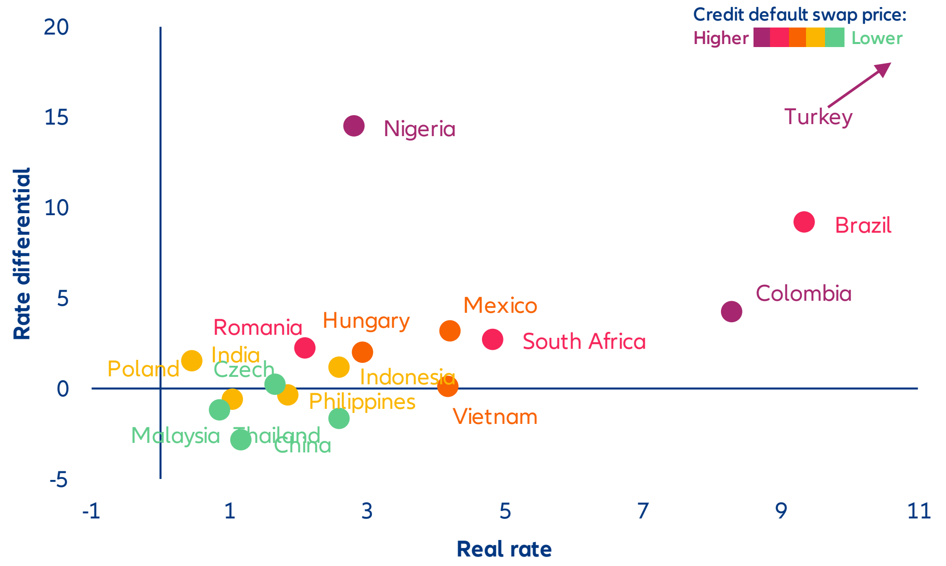


Note: International reserves including gold

Sources: LSEG, Allianz Research

In this context, the Middle East escalation arrives at a positive moment for rate differentials as the US benchmark Treasury saw a broad fall, defined by a structural repricing of the US fiscal risk premium during 2025. During 2025, real rates improved as inflation across EMs decelerated faster than nominal rates, allowing central banks to preserve a meaningful real rate buffer without undermining currency stability or triggering disruptive capital outflows. This parallel configuration revealed a broad improvement the EM universe while highlighting clear groupings within the universe. Türkiye, Brazil, Colombia, Nigeria, stand out with the widest nominal differentials to UST, though the investment case differs substantially across these markets. Türkiye, Brazil and Colombia also offer elevated real rates, creating a rare alignment of internal and external value. South Africa, together with Mexico sit in an intermediate position, with an elevated differential and a high real rate, though less extreme than Brazil and Colombia. Meanwhile, China, Thailand and Malaysia offer small rate differentials to the US, suggesting these markets are priced primarily for domestic investors.

Figure 6: Real rate, rate differential and credit default swaps in selected EMs



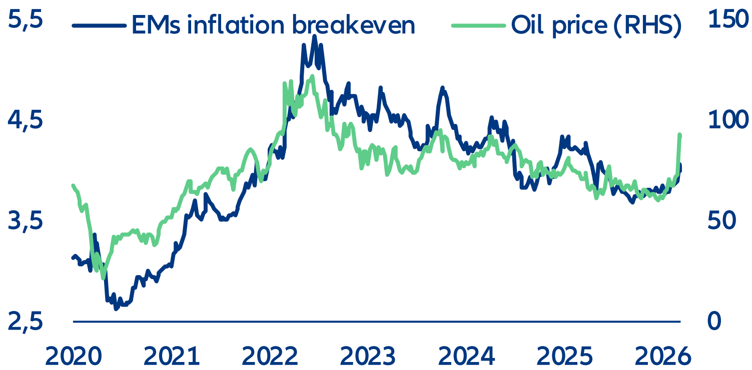
Sources: LSEG Workspace, Allianz Research

However, this new geopolitical shock renews stagflationary fears, with central banks on hold due to higher inflationary pressures. Inflation expectations have surged given the quick increase in prices during the one week of closure of the Strait of Hormuz – a dynamic already visible in US Treasury markets, where yields have risen sharply rather than fallen, as bond markets price the conflict primarily as an inflation shock rather than a growth one. This puts central banks in an acutely uncomfortable position: slowing growth would ordinarily call for rate cuts, but persistent energy-driven inflation prevents them from acting. The Fed, which had been on a gradual easing path, could now change path and keep on hold, or even hike through the rest of 2026. For emerging market central banks – many of which, like Egypt and Türkiye, are already navigating fragile disinflation paths – the constraint is even more binding, as currency depreciation pressures compound imported inflation, leaving little room to support growth without risking a renewed inflation spiral.

Emerging-market sovereign spreads have widened modestly, reflecting the tone of cautious sentiment. The move has been more pronounced across Africa, the Middle East and CEE Europe. The sharpest adjustments have occurred in the most directly exposed countries, such as Bahrain or Pakistan, where acute energy-import vulnerabilities have intensified market sensitivity. In contrast, the rest of the Gulf has remained relatively tight, supported by strong oil-related revenues that partially offset elevated geopolitical risks. This resilience should nonetheless be interpreted carefully. Historically, spread markets have tended to reprice geopolitical shocks with a delay, responding more forcefully once hard economic data begins to reveal the underlying impact. Should the Strait of Hormuz remain closed beyond roughly four weeks, and supply disruptions start feeding into macro indicators, a broader and more disorderly widening across EM credit, particularly among energy-import-dependent sovereigns, cannot be ruled out.

The critical question is whether local bond markets are beginning to price the inflation risk. Breakeven inflation (BEI) offers a real-time measure of market-implied inflation expectations. Since oil prices are an important driver of inflation compensation (Figure 7), a rise in oil should normally push EM breakevens higher. However, before the strikes, inflation protection across EM looked unusually cheap. For a representative basket of economies, the average 10-year breakeven stood at just 3.89%, while the historical sensitivity to a 10% rise in oil prices averaged only +8bp in BEI – well below what our simulations suggest on actual inflation rate. In other words, inflation expectations were still responding to oil, but much less than in past stress episodes. This matters because the sensitivity of BEI to oil tends to rise when markets start to view the shock as more persistent and structural. If the conflict proves prolonged, inflation expectations across EM curves are therefore likely to adjust more forcefully.

Figure 7: EM 10Y inflation breakeven (excl. Türkiye) and oil price

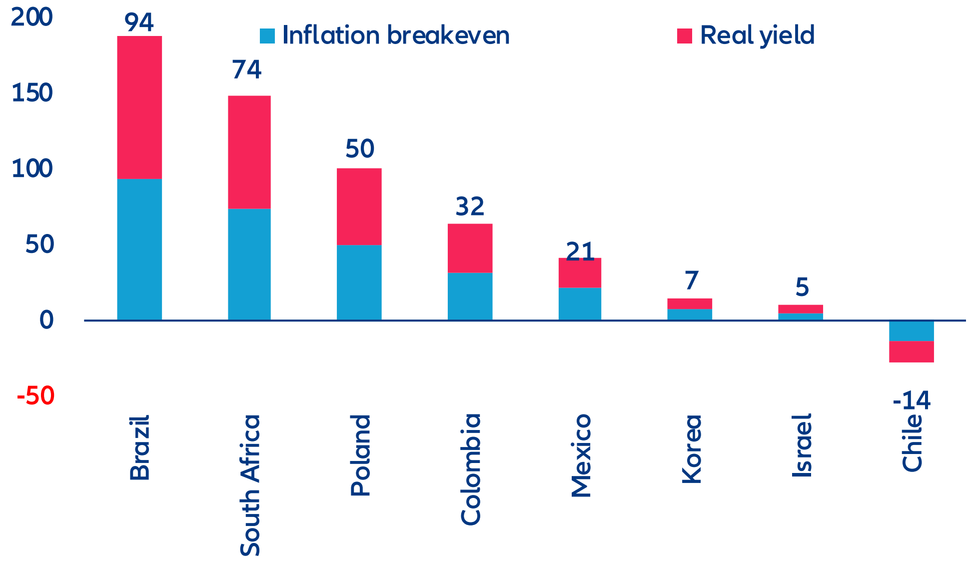


Sources: Bloomberg, LSEG Datastream, Allianz Research. Notes: Inflation breakeven is derived only for countries that have an active inflation-linked bond market. Countries considered are Brazil, South Africa, Poland, Mexico, South Korea, Türkiye, Chile, Israel and Colombia.

Has that adjustment already begun? Early evidence suggests that repricing is underway, but not uniformly. In the first week after the strikes (27 February to 9March), South Africa recorded the largest breakeven widening, at +57bps, from 4.32% to 4.89%, consistent with its current-account vulnerability and energy import dependence. Poland's breakeven rose by +34bps and Mexico's by +28bps – meaningful moves in markets where pre-shock inflation expectations were running close to, or even below, central bank targets. However, the adjustment has not been universal. Colombia's breakeven fell by 45bps, as markets priced in the terms-of-trade gains of a net oil exporter, while Israel's rose only +10bps.

A look at nominal 10-year yield moves helps clarify the forces behind the repricing and the extent to which it reflects inflation versus broader risk aversion (Figure 8). Chile now stands out as the clearest case of inflation-led repricing: its +18bps rise in nominal yields was more than explained by a +33bps widening in breakevens, while real yields fell by 15bps, indicating that the move was entirely driven by higher inflation expectations. Turkey (+233bps nominal, including +181bps from breakevens) and Korea (+26bps, of which +20bps came from breakevens) also saw most of their repricing driven by inflation compensation. Mexico, which initially looked almost entirely inflation-driven, has since experienced a broader sell-off: nominal yields are now up +49bps, with +28bps attributable to breakevens and +21bps to real yields. This still points to an inflation-led move, but with a more visible risk-premium component than before. Brazil presents a much sharper contrast. Nominal yields rose by +88bps even as breakevens fell by 6bps, implying a +94bps increase in real yields and pointing clearly to higher credit and liquidity premia rather than inflation. South Africa has also moved in that direction. With nominal yields now up +96bps, breakeven widening has largely reversed to just +22bps, leaving +74bps accounted for by higher real yields, suggesting that the move is now dominated by risk aversion rather than the inflation repricing seen initially. Colombia (-13bps) remains the outlier, with nominal yields still rallying as markets continue to price terms-of-trade gains for a net oil exporter, while Israel (+14bps) has shifted from a modest rally to a mild sell-off. In CEE, Türkiye, Romania, and Hungary have recorded the sharpest sell-offs in local rates markets, with Poland not far behind. Türkiye's move (+233bps) – the most pronounced across all EM – reflects a confluence of acute oil import dependency, geographic proximity to the conflict, and a market that was already pricing fragile disinflation dynamics. Romania (+101bps) has overtaken Hungary (+94bps) as the second-largest mover in the region: both had benefitted from a strong rates rally going into the conflict, and the sell-off is in part a technical correction as investors take profit on stretched long positions, compounded by the region's energy vulnerability and broad CEE risk-off repricing. Poland (+83bps) and the Czech Republic (+59bps) round out a picture of broad-based CEE weakness.

Figure 8: EM 10Y local yield moves after the conflict: inflation breakeven vs real yield contribution, bps



Source: Bloomberg, LSEG Datastream, Allianz Research. Notes: Changes in yield from 27/02/2026 to 13/03/2026.

The outlook for EM capital flows in 2026 is now firmly scenario-dependent. In a short-lived conflict – contained within four to five weeks – EM inflows should remain broadly positive, with markets normalizing relatively quickly toward pre-Liberation Day conditions. Investors would nevertheless become more selective, differentiating more sharply between countries on the basis of energy exposure and inflation vulnerability rather than treating EMs as a homogeneous asset class. A prolonged conflict would alter that picture materially. Flows would rotate toward safer assets, while the most exposed economies – Egypt and Pakistan foremost among them – would face the sharpest outflow pressure as the combination of currency stress and imported inflation reshapes their risk profile. In this environment, the repricing is likely to take the form of a selective energy risk premium, differentiating sharply between exposed and insulated sovereign issuers rather than triggering a broad-based EM sell-off.

This scenario dependency comes against the backdrop of an exceptionally strong year for EM debt issuance. On the hard-currency side, sovereign issuance rebounded from roughly USD72bn in 2022 to around USD236bn in 2025 – the highest level on record – reflecting genuinely renewed access to international funding markets. Saudi Arabia and Türkiye were the primary drivers, with Türkiye's issuance alone rising from USD12bn in 2022 to roughly USD31bn in 2025. Poland also contributed significantly, with issuance climbing to around USD30bn in 2024, while the UAE and Mexico provided additional support. Local-currency issuance expanded in parallel, led by Brazil, Poland and India, with the Czech Republic emerging as a particularly notable issuer at USD195bn – the largest absolute increase over the period – further supported by Thailand and Mexico. The scale of this recovery underscores how much is at stake: a prolonged conflict risks interrupting a multi-year rebuilding of EM market access at precisely the moment it had regained its strongest footing in years.

Slightly more than two weeks into the conflict, we observe modest underperformance among the more exposed sovereign issuers. While the Philippines continues to trade broadly in line with the wider market, Egypt – and in particular Türkiye and India – have experienced more pronounced spread widening. If the Ukraine conflict is taken as a reference point, spreads could widen by roughly one-third to one-half of pre-crisis levels. In practical terms, this would imply an estimated widening of around 80bps for Türkiye, 35bps for the Philippines, 180bps for Egypt and 35bps for India. It is worth noting, however, that Türkiye's macro-financial position is materially stronger today than in early 2022, and that energy-related risk premia during the Ukraine episode ultimately proved short-lived.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed
or implied in such forward-looking statements.
Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends,
(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

The company assumes no obligation to update any information or forward-looking statement contained herein,
save for any information required to be disclosed by law.

Allianz Trade is the trademark used to designate a range of services provided by Euler Hermes.

Allianz Research | 16 March 2026

Warsh's double dilemma: when the Middle East rewrites the Fed's playbook

Ludovic Subran
Chief Investment Officer & Chief Economist
ludovic.subran@allianz.com

Maxime Darmet
Senior Economist US-UK-France
maxime.darmet@allianz-trade.com

Patrick Krizan
Senior Rates Strategist
patrick.krizan@allianz.com

Dorian Simon
Rates Strategist
simon.dorian@allianz.com

In Summary

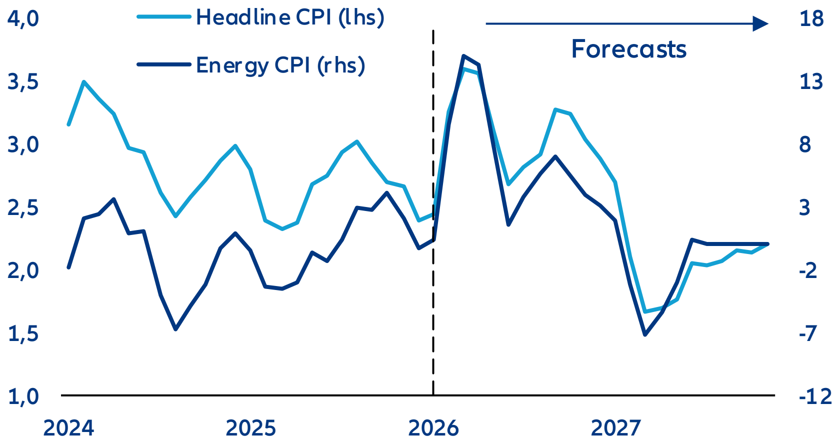
- **The energy price shock will delay the Fed's single rate cut in 2026.** Rising energy prices should push US inflation towards +3.6% y/y in April-May, up from +2.8% expected before the Middle East conflict, under the assumption that oil prices remain around 90 USD/bbl on average in Q2. Despite a still weak labor market, the Fed will have to keep rates on hold at least through the summer amid risks of de-anchoring inflation expectations. We continue to expect only one 25bps rate cut this year, but pushed out to September.
- **Fed chair nominee Kevin Warsh combines a dovish view on interest rates with a hawkish approach to the Fed's balance sheet, which could drain liquidity from a highly leveraged financial system. Increasing frictions in the US money market would have a global impact on financial markets mainly though the channels of FX and interest-rate volatility.** Warsh opposed additional rounds of quantitative easing (QE) in the 2010s, warning that prolonged asset purchases fueled financial excess and risked inflation. With hindsight, his concerns about loose financial conditions and asset valuations appear partly validated. But reducing the Fed balance sheet drastically or even returning to a scarce-reserves regime would ignore the intermediation mechanisms on which the US funding market has been founded since the GFC. The critical point is not the amount of reserves but the dealers' balance sheets available to warehouse Treasuries and intermediate repo flows. Dealers face regulatory limits that are capping the size of their balance sheets. They cannot do both: absorb heavy Treasury issuance and provide smooth funding for the private sector. It would drive up prices for hedging instruments (swap spreads and cross-currency basis) but in a worst-case scenario such mismatch could trigger a deleveraging spiral affecting valuations of risky assets and increasing default risk for highly leveraged entities (e.g. hedge funds). The Treasury and the Fed have options to mitigate this, but it would require a massive easing of leverage regulation and/or a substantial decrease in Treasury issuance, the first being more likely than the latter.
- **We expect the Fed to move cautiously towards a modestly smaller balance sheet from Q4 2026 at the earliest, keeping Treasury holdings steady while continuing the run-down of the USD2trn MBS holdings. This would limit money-market volatility but still risk higher mortgage rates. Wars typically involve monetizing deficits, i.e. restarting QE and departing from Warsh's pledges for rapid quantitative tightening (QT).** The Fed will likely adopt a hybrid model: a leaner balance sheet combined with standing repo facilities as permanent liquidity backstops, preserving rate control while mitigating systemic funding risks. Regulatory moves on bank liquidity regulations could also support the transition toward lower bank's reserves. We would expect banks' reserves balances to drop gently from 9.5% of GDP currently to reach the level of the 2019 money-market meltdown (below 7% GDP) only by Q4 2028, though uncertainties are large and episodes of money market volatility will likely become more frequent. The MBS reduction would however blunt the administration's attempts to lower mortgage rates. The conflict in the Middle East could also drastically change the course of monetary policy in the US. In a downside scenario, quantitative easing may be needed to help finance the effort of the war.

The energy price shock will delay the Fed's single rate cut in 2026

The Fed is likely to keep interest rates on hold at its next meeting on 18 March amid upward pressure on inflation, which we expect to peak at +3.6% by April/May. The conflict in the Middle East poses a difficult dilemma for the Fed, with inflation set to pick up amid higher energy prices while the labor market still looks fragile. Non-farm payrolls declined by -92K in February, and though part of the weakness was due to strikes in the healthcare sector, the private sector excluding healthcare continued to shed jobs. But with inflation now stuck above the 2% target for five years, the Fed clearly does not have the luxury to focus on labor market risks. Inflation expectations are especially sensitive to energy (and food) prices and could risk de-anchoring if the Fed is perceived as too complacent with inflation risks. We expect the Strait of Hormuz to open again by the end of April. In the meantime, partial re-retouring, additional production (Russia, US) and tapping strategic reserves should cover over 50% of the gap in oil and gas, capping oil prices at 90 USD/bbl on average in Q2 2026. We estimate that US energy CPI will rise from +0.4% year-on year in February to peak at around +15% in April. Headline inflation would rise from +2.4% to a peak of +3.6% in April-May (Figure 1).

We continue to expect one 25bps rate cut this year, but pushed out to September. We have long argued that the Fed will be limited in its ability to cut interest rates in 2026, and we were expecting only rate cut in June prior to the conflict in the Middle East – a much more hawkish view than what is currently priced in financial markets. We now expect the Fed to wait until September, when inflation should start to decline though still remain above target. The labor market is likely to remain weak, or even weaker further, through the summer as higher inflation eats into households' incomes and some corporates' profit margins. If the Fed stays on hold in the near term, the risks of second-round effects on prices, which would compound the initial energy price shock, may be smaller than in 2022.

Figure 1: US CPI inflation forecast (left)



Sources: LSEG Datastream, Allianz Research

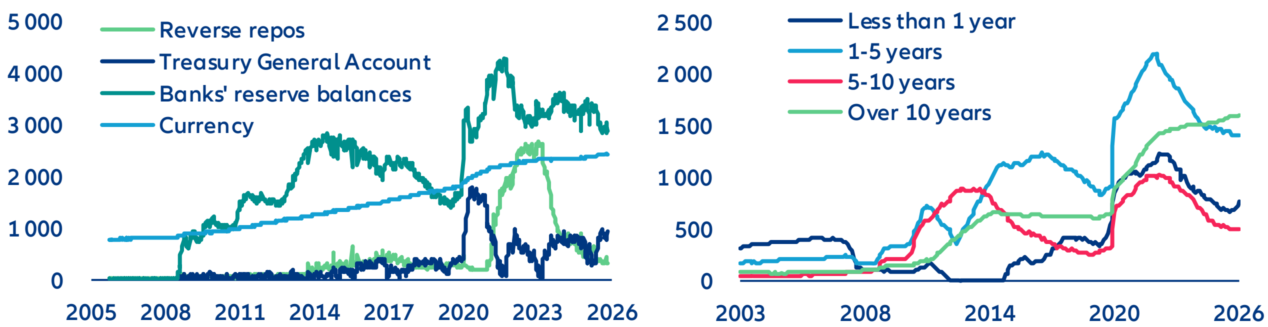
Kevin Warsh' s push for a lower Fed's balance sheet could gain traction…

Kevin Warsh, President Trump's nominee for the next Fed Chair, has assuaged financial markets, thanks to his strong credentials and perceived prudent approach on monetary policy. Warsh, who is expected to take over in May, served on the Fed Board from 2006 to 2011 during the height of the Global Financial Crisis (GFC). At the time, he supported an aggressive monetary response to stabilize markets. However, once the recovery gained traction, he opposed the second and third rounds of quantitative easing (QE) and stepped down in 2011, reportedly amid policy disagreements over continued asset purchases. Importantly, his concerns about QE were centered on the blurring of monetary and fiscal policy. He has warned about financial stability risks, asset bubbles and inflation pressures stemming from an expanded balance sheet. With the benefit of hindsight, many of Warsh's views proved to be well founded: Financial conditions have been very loose since the aftermath of the GFC, equity markets have been buoyant, financial risks have built up and inflation picked up – though at a much later stage. Other critics of the Fed's balance sheet policy include Treasury Secretary Bessent, whose line of attack is on the net operating losses that the Fed makes by paying interest on banks' reserves – although Bessent is not pushing for a sharp shrinkage of the Fed's balance sheet. More recently, Warsh argued in op-ed in 2025 for a significant reduction in the Fed's holdings.

However, Warsh's hawkish views on the balance sheet sit alongside more dovish views on policy rates. Warsh argues that money is too tight for small corporates and calls for lower policy rates. He believes strong GDP growth is being driven by a positive supply shock – notably AI – which should generate disinflationary forces, which will in turn be reinforced by the Administration's deregulatory agenda. Besides, in 2025, Warsh wrote that the Fed "should abandon the dogma that inflation is caused when the economy grows too much, and workers get paid too much," signaling limited concern about demand-driven inflation pressures. He also suggested that balance sheet "largesse can be redeployed in the form of lower interest rates to support households and small and medium-size businesses." While the operational implications remain unclear, this could imply a preference for shifting from asset purchases toward more direct credit-support mechanisms.

Warsh's criticism of the Fed's bloated balance sheet could find support among FOMC members. The Fed has already shrunk its balance sheet significantly from its 2022 high of 35% of GDP to less than 20% through a reduction of asset holdings (Treasuries and MBS). All Treasuries maturities were reduced except long-term ones (Figure 2, left). The pre-GFC average of the Fed's balance were close to 5% of GDP, when the Fed operated under a ''scare" reserve system rather than an ''excess'' reserves one: banks' reserve balances were essentially zero prior the GFC (Figure 1, right). In end-2025, following tensions in the repo market, the Fed decided to launch Reserve Management Purchases (RMPs), i.e. Treasury purchases at short maturities (T-bills) to increase reserve balances again. The T-bills purchases are partly funded by the reinvestments of MBS and longer-dated Treasuries paydowns. However, there was some disagreement among FOMC members around the extent to which the Fed should increase its purchases, with the minutes noting "a couple of participants added that effective standing repo operations may allow for a smaller balance sheet ".

Figure 2: Fed's balance sheet main liabilities items (left) ; Fed's Treasury holdings by maturity (right) – in USD bn

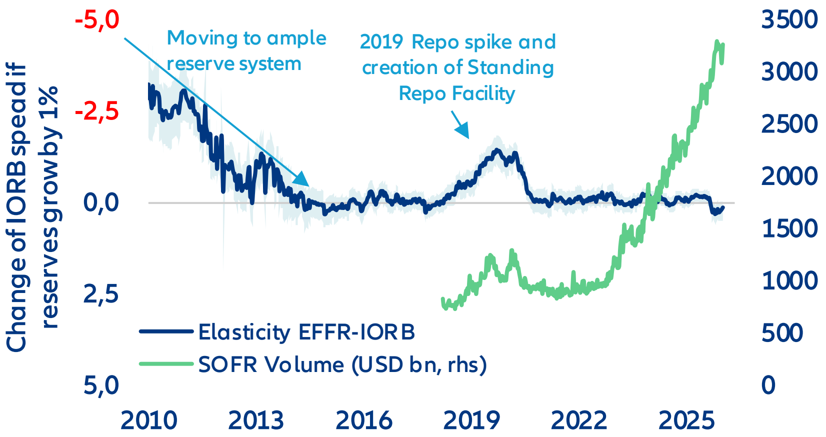


Sources: LSEG Datastream, Allianz Research

...but it would be a risky strategy for little return

The transition towards a reduced balance sheet would entail risks for financial stability as the monetary policy system has shifted from reserve fine-tuning to private-sector intermediation. The current ample reserve balance sheet policy is the result of the structural breaks of the GFC. Before 2007, the Fed ran a scarce-reserves corridor system in a funding market structured by large scale unsecured lending, small dealer balance sheets, loose leverage constraints (pre-Basel III), off balance-sheet repo and a limited role of non-banks (money market funds, hedge funds etc.) in cash supply. After the liquidity collapse during the GFC, the Fed expanded its balance sheet by large-scale asset purchases – creating such excess reserves that the corridor system could no longer be handled with fine-tuned open-market operations – and endorsed the role of market maker of last resort. The US funding market structure shifted from one where banks arbitraged small reserve shortages via interbank lending to one where administered rates and central-bank facilities anchor pricing and balance-sheet-constrained dealers intermediate between non-banks and the Fed. Consequently, the critical point of the system shifted away from the amount of reserves to the amount of dealer balance sheet available to warehouse Treasuries and intermediate repo flows. If balance-sheet capacity is constrained, market liquidity can be scarce despite ample reserves. Going back to a pre-GFC scarce reserve system would not be "doing what worked for decades" but simply ignoring the intermediation nature of the USD funding market. The current ample-reserves floor system with Interest on Reserve Balance (IORB) and standing facilities (Overnight Reverse Repo ON RRP, Standing repo facility SRF) has been a crucial pillar of central banks' volatility compression in the post-GFC era. It has immunized the monetary policy system against shifts in the reserve demand. Changes in reserves today only have a minimal impact on movements in money-market spreads. Under the previous "scarce reserves" regime, a 1% change in reserves could trigger a response of up to 3–4 bps in the money-market spreads (Figure 3). A return to scarce reserves with daily open market operations would create volatility and money-market rate dislocations as rather than simple management of reserves, intermediation capacity is the key point of friction.

Figure 3: Reserve demand elasticity: EFFR* - IORB spread reaction to 1% change in reserves



**EFFR = Effective Fed Funds Rate*

Reserve Demand Elasticity is the responsiveness of money-market spreads (Effective Fed Funds-IORB, SOFR-IORB) to changes in reserve supply, i.e. the slope of the reserve demand curve. When frictions are low, this slope is near zero and spreads hardly move. When transmission is impaired the curve steepens and small reserve shocks lead to large changes in money market spreads. In 2009, a 1% expansion of reserves led to 3-4 bps contraction of money market spreads, versus only 0.15bp today. SOFR-IORB elasticity is more volatile as it captures the transmission between unsecured and collateralized money market rates.

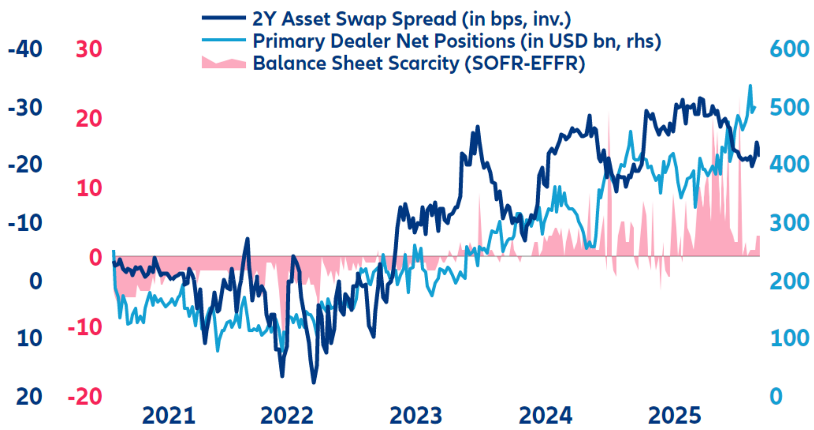
Sources: LSEG Datastream, Allianz Research

A removal of the Interest on Reserve Balance (IORB) would have more negative effects than positive ones – making it unlikely. Alternatively to the Fed balance sheet discussion, there is an ongoing debate on removing the Interest On Reserve Balances. (IORB). The IORB is intended to steer banks' behavior by guaranteeing an administered return on their reserves. The costs and benefits of operating this system are politically challenged. Critics see the IORB as a subsidy for the US banking sector. The Fed sees the IORB as the price to sustain bank credit flow. Eliminating the IORB "would be extraordinarily disruptive" as it would impair the Fed's control of short-term rates, reduce the volume of bank-intermediated transactions, and thereby diminish private-sector liquidity. In addition it would generate massive money market volatility and downward pressure on short-term rates as banks would substitute reserves with interest-bearing alternatives such as newly issued Treasuries or the Fed's reverse repo facility (RRP, which is the lower bound of the policy rate corridor). Replacing reserves with Treasuries would increase banks' liquidity risk and raise Basel III capital charges, boosting banks' demand for central bank liquidity and therefore counteracting efforts to shrink the Fed's balance sheet. Shifting to the RRP would also compress banks' net interest margins, reduce their lending capacity and restraining credit provision to the economy. Given these adverse consequences, the costs of a IORB removal outweigh fiscal benefits.

The Fed is already operating in a framework of scarcity

The Fed is facing a funding paradox where an ample balance sheet coincides with scarce intermediation capacity. While the discussion is focused on the scarcity of reserves, we find that the Fed is already operating in a framework of scarcity but from the intermediation side. Intermediation in funding markets is the willingness and ability of balance-sheet-constrained dealers and banks (subject to regulatory leverage and liquidity limits) to act as a broker between cash-rich non-banks (money market funds, pension funds etc.) and collateral-rich issuers (US Treasury, corporates etc.). During the QE era, balance-sheet constraints were rarely the binding margin. With QT, excess liquidity declined and dealers' balance-sheet capacity decreased as did their debt-absorption ability. This situation prevails today. With Treasury (notably T-Bills) issuance taking over an increasing part of the scarce dealers' balance sheet, funding becomes more expensive for other investors, especially those running leveraged risky strategies (i.e. basis trade, asset swap spread). In Figure 4, we can see the effect on balance sheet scarcity: the more Treasuries issuance increases, the more dealers must increase their holdings on their books. Their cost of carry is reflected in the risk premium, which is the asset swap spread, and in the increased demand for cash reflected by the SOFR-Fed Funds spread (SOFR can be understood as the price for balance-sheet capacity) as well as volatility in short term rates.

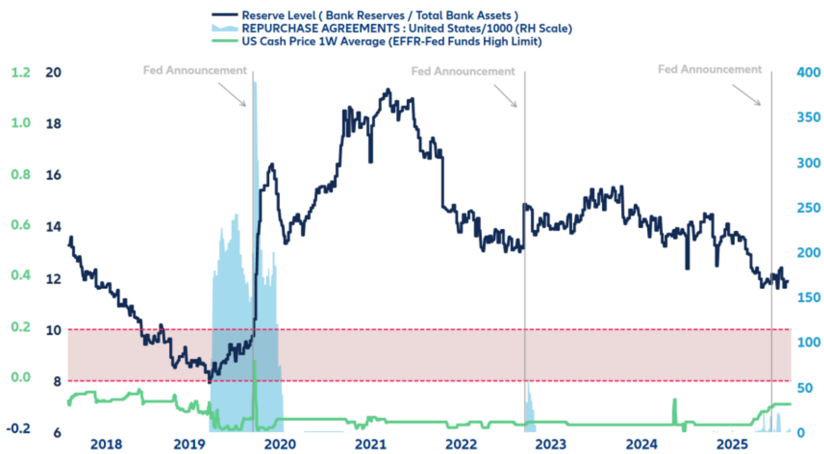
Figure 4: Balance-sheet scarcity illustrated



Sources: LSEG Datastream, Allianz Research

The recent decision of the Fed to resume buying short-term US Treasuries (T-Bills) must be seen as a reaction to this scarcity, acting as cash support to absorb heavy US Treasury issuance. If intermediation scarcity prevails we will see further Fed interventions mostly though the SRF (Temporary Open Market Operations) and further asset purchases. The probability of Fed interventions is linked to the ratio of bank reserves on bank total assets. Above 10% reserves are "abundant", below 10% reserves are "ample" while 8% marks reserves as "scarce". As shown in Figure 5, when the market reaches the level of absorption stress the Fed generally intervenes and restores market liquidity. These interventions are often accompanied by announcements of new facilities (SRF in September 2019) or programs (Bank Term Funding Program following Silicon Valley Bank Crisis, March 2023).

Figure 5: The Fed intervention framework - cash price, reserve levels and standing Repo facility use



Sources: LSEG Datastream, Allianz Research

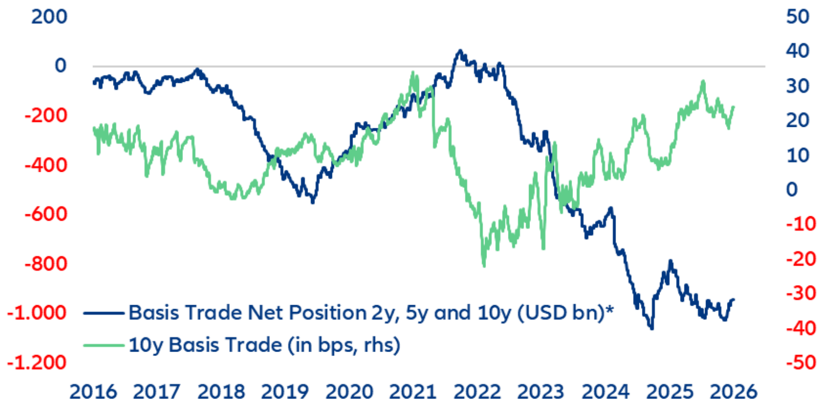
The Fed is moving from being a structural provider of liquidity to a relief valve while the balance sheet is playing a limited role. The problem is that non-banks (esp. leveraged hedge funds) are not "plumbed" to the Fed's valves.

Elevated and volatile repo rates with tighter haircuts can force them to unwind risky leveraged strategies (such as basis trades) and sell Treasuries into less liquid market, amplifying price moves and volatility. Such a destabilizing deleveraging spiral was seen in March 2020, for instance. The Treasury and the Fed have option to mitigate these risks. A substantial easing of leverage regulation (i.e. SLR, ISLR) could alleviate some balance-sheet constrains and free up cash that could ease the liquidity pressure. A substantial slowdown in Treasury issuance could also free balance-sheet capacity, but remains unlikely. Both would anyway only be temporary.

The Fed could lose its dampening grip on global volatility

US money markets are the primary providers of liquidity and leverage across the global financial system. When US money-market spreads widen and volatility rises due to pressure on SOFR, the FX swap market and interest-rate swaps become the main transmission channels. For FX swaps this occurs because the same intermediaries use their constrained balance sheets to price FX derivatives. Therefore intermediaries can pass on the cost of scarce balance sheets through higher risk premia onto the end users, resulting in higher FX volatility and widening the cross-currency basis, appreciating the USD, which tightens global financial conditions. The transmission channel through interest-rate swaps operates via Treasury issuance dynamics. Swap spreads reflect liquidity at specific maturities as well as the banking system's absorption capacity. If newly issued debt is not enough subscribed by the private sector, primary dealers (banks) must by law keep it on their books, affecting their funding capacity of the private sector. Negative swap spreads are a sign that issuance absorption is tight. This reduces the demand for duration, increases bid-ask spreads, tightens trading volumes and steepens the curve. This effect is reinforced by huge positions of hedge funds in bond basis trades (long cash bond, short future) funded in US money markets. Official numbers from the CFTC (Commodity Futures Trading Commission) indicate a total volume of USD1,300bn with a USD1,000bn net short position in the US basis trade (Figure 6).

Figure 6: Hedge funds have a minimum net short position of USD1000bn in US Treasury futures

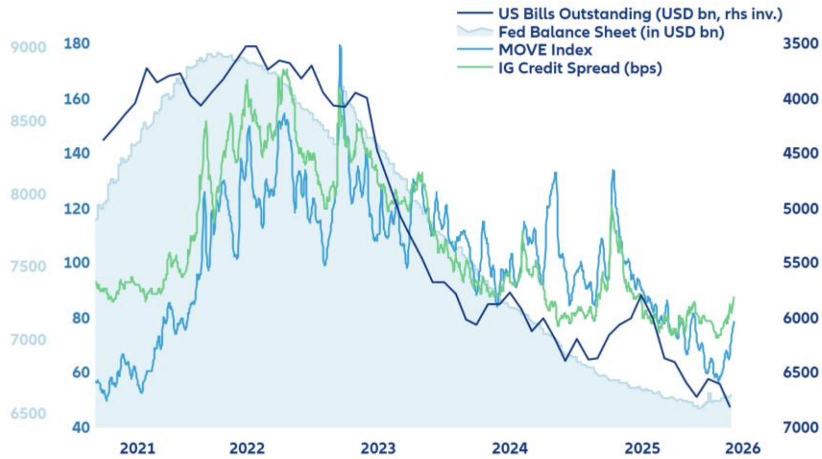


**net open interest in USD for leveraged funds as reported by the CTFC (Commodity Futures Trading Commission)*

Sources: LSEG Datastream, Allianz Research

Scarce money market liquidity therefore creates a risk of fixed income market selloff and increases the risk of a massive deleveraging spiral. The US Treasury market is now more exposed to this risk as hedge funds have become the marginal buyer of US Treasuries since the of the US curve (on a FX hedged basis) has become relatively unattractive for foreigners. But hedge funds are not buy-and-hold investors; they use these US Treasuries as collateral to borrow cash in the repo market to fund leveraged positions such as yield curve arbitrage, swap spreads, credit spreads, securitized products on bonds, crypto or stock indices. This demand from leveraged investors compresses implied volatility on many asset classes and is a reason for the unusually low global volatility. But this mechanism only works as long as there is subdued volatility on money-market rates. This is why any change in the monetary policy system by the Fed would be very risky (Figure 7).

Figure 7: Volatility compression is linked to monetary and fiscal policy



Sources: LSEG Datastream, Allianz Research

How would a global volatility shock unfold?

The transmission of rising money-market rates volatility runs via higher risk premia that would hurt leverage, affect risky-asset valuations and increase default risk within the banking and non-bank system. It would unfold the following way: First, a margin call due to rates volatility would trigger emergency refunding demand and so a selloff in leveraged assets. Fringe assets would be the first in line to correct (leveraged crypto, highly leveraged ETFs). Second, default risk among leveraged investors (hedge funds) would rise. Since hedge funds usually fund themselves on the sponsored repo market[1], the sponsoring banks would finally have to step in and bail out the clearing house if hedge funds fail. This procyclicality is the result of the "Active Treasury Issuance" (ATI) strategy. Its objective is to massively issue T-bills to release excess liquidity created by QE and parked into the RRP (reverse repo program) to counter restrictive effects from Quantitative Tightening (QT). But this aggressive supply of T-Bills also increased the potential for leveraged trades for non-banks (hedge funds). ATI therefore contributed to fuel asset prices and compresses asset volatility. So far we do not see stress on sponsored repo, but outstanding volumes indicate a slight deleveraging trend since the beginning of the year. The deleverage risk from the non-bank sector is reinforced by additional layers of leverage: collateral transformation and collateral re-use. Collateral transformation refers to the practice where counterparties swap lower-quality assets, such as investment-grade corporate bonds, for high-quality assets like US Treasuries in uncleared bilateral markets. These transactions occur off-balance sheet, their outstanding amount is hard to quantify. Collateral re-use (rehypothecation) occurs when dealers re-use received collateral forming collateral chains in which the same asset runs through multiple transactions. This is a widely used practice as around 85% of collateral received by the biggest primary dealers is said to be re-used. In a massive sell-off scenario, collateral re-use might trigger a chain reaction across markets that the Fed liquidity back stop could only partially contain as it is occurs outside its sphere of influence in the non-bank and/or uncleared segment of USD money markets (Eurodollar system).

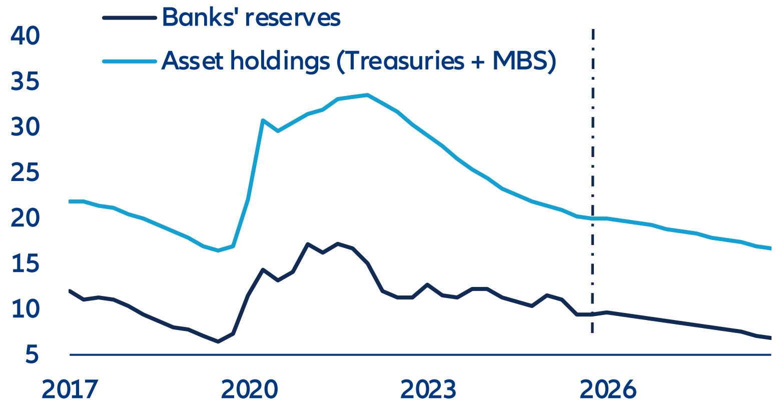
The Fed will reduce its balance sheet in slow-motion

Given these market fragilities, we expect the Fed to embark cautiously on a balance-sheet reduction path, from Q4 2026 at the earliest. Treasury holdings could be maintained at flat levels, while the Fed could continue the run-down of the USD2trn MBS holdings. The Fed would then instead be relying more heavily on standing repo facilities as permanent liquidity backstops, preserving rate control while mitigating systemic funding risks while reserves decline. Regulatory moves on bank liquidity regulations, such as a lowering of liquidity requirements for banks, which include banks' reserves, could support the transition towards a smaller balance sheet and lower reserves. This would maintain financial market stability but risk higher mortgage rates. We would expect it would

[1] Sponsored repo market refers to cleared repo transactions in which a dealer (bank) sponsors non-dealer clients (such as hedge funds or money market funds) into the Fixed Income Clearing Corporation's central clearing platform, allowing those clients indirect access to centrally cleared repo while the dealer guarantees their obligations.

take a bit of time for Warsh to convince a majority of the FOMC to start reducing the Fed's balance sheet. Most likely, a compromise could be found on flattening total Treasuries holdings, rather than shrinking them, but to let the MBS run-down continue at the current pace of around -USD17bn per month. In this baseline scenario, we would expect the Fed to start shrinking its balance sheet in Q4 2026. Banks' reserves would drop from 9.5% GDP currently to less than 7% in Q4-2028 – a level close to the 2019 money-market meltdown. Put otherwise, under prudent balance-sheet reduction, it would take a bit of time before the Fed potentially faces money-market volatility on the back of scarcer reserves. However, forecasting is fraught with large uncertainties. In particular, should nominal GDP grow faster than we expect, we could see a faster reduction of reserves as a share of GDP.

Figure 8: Banks' reserves at the Fed & Fed's asset holdings (Treasuries + MBS), % GDP



Sources: LSEG Datastream, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed
or implied in such forward-looking statements.
Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends,
(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

The company assumes no obligation to update any information or forward-looking statement contained herein,
save for any information required to be disclosed by law.

Allianz Trade is the trademark used to designate a range of services provided by Euler Hermes.